

Limit of a function, group *Alpha* α -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1} \frac{2}{(4x-4)^3}$??? does exist

(b) $\lim_{x \rightarrow \infty} 3^{-1-5x^2}$??? converge

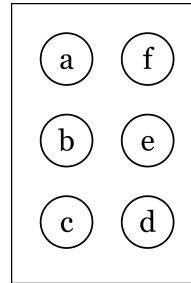
(c) $\lim_{x \rightarrow \infty} \left(7x - \sqrt{49x^2 - 3x - 3}\right)$??? $-\frac{3}{7}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x-4}{-5x-6}\right)^{x/9}$... ??? ... $e^{-2/45}$

(e) $\lim_{x \rightarrow -2} \frac{-2x^2 - 11x - 14}{x^2 - 4}$... ??? ... $\frac{3}{4}$

(f) $\lim_{x \rightarrow 1} \frac{\sqrt{6} - \sqrt{x+5}}{6x-6}$??? $\frac{-1}{12\sqrt{6}}$

1.



Braille letter

Limit of a function, group *Alpha* α -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 5/2} \frac{-1}{(2x-5)^{15}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 5^{4-8x^2}$??? converge

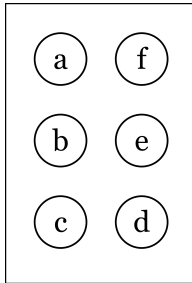
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{36x^2 - 7x - 3} - \sqrt{36x^2 + 9x - 4}\right)$??? $\frac{1}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x-5}{-2x-4}\right)^{x/7}$??? $e^{9/14}$

(e) $\lim_{x \rightarrow -3} \frac{-4x^2 - 11x + 3}{x^3 + 3x^2 - 6x - 18}$??? 0

(f) $\lim_{x \rightarrow -6} \frac{1 - \sqrt{x+7}}{2x+12}$??? $-\frac{1}{4}$

2.



Braille letter

Limit of a function, group *Alpha* α -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 4/3} \frac{5}{(-3x+4)^6}$.. ??? .. does exist

(b) $\lim_{x \rightarrow \infty} 2^{-3-2x^2}$??? diverge

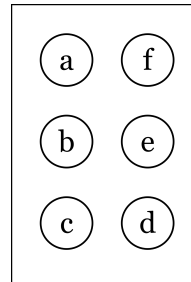
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 - 2x - 5}\right)$.. ??? .. 1

(d) $\lim_{x \rightarrow \infty} \left(\frac{-4x-5}{-4x-3}\right)^{x/5}$??? ∞

(e) $\lim_{x \rightarrow -5} \frac{-5x^2 - 29x - 20}{2x^3 + 10x^2 - 3x - 15}$. ??? . $\frac{21}{47}$

(f) $\lim_{x \rightarrow -6/5} \frac{2x + \frac{12}{5}}{\sqrt{x+3} - \sqrt{-3-4x}}$??? $\frac{12}{\sqrt{125}}$

3.



Braille letter

Limit of a function, group *Alpha* α -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -4} \frac{4}{(x+4)^8}$??? does exist

(b) $\lim_{x \rightarrow \infty} 3^{1+2x^2}$??? converge

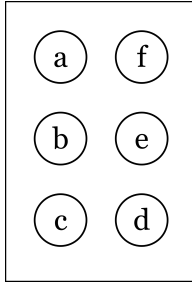
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{4x^2 - 2x + 5} - \sqrt{4x^2 + 3x - 4}\right)$??? $\frac{1}{4}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{4x-3}{4x+3}\right)^{x/4}$??? ∞

(e) $\lim_{x \rightarrow -2} \frac{2x^2 + 6x + 4}{x^2 - 4}$??? $\frac{1}{2}$

(f) $\lim_{x \rightarrow 2} \frac{\sqrt{5} - \sqrt{7x-9}}{9x-18}$??? $\frac{-7}{18\sqrt{5}}$

4.



Braille letter

Limit of a function, group *Beta* β -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1/6} \frac{8}{(-6x+1)^3} \dots ??? \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 6^{4+x^2} \dots ??? \dots$ converge

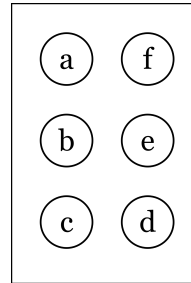
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x - 1} - \sqrt{x^2 - 5x + 2} \right) ??? 6$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x-5}{-2x-6} \right)^{x/6} \dots ??? \dots e^{11/12}$

(e) $\lim_{x \rightarrow -1} \frac{3x^2 + 10x + 7}{x^2 - 1} \dots ??? \dots -\frac{2}{3}$

(f) $\lim_{x \rightarrow 2} \frac{\sqrt{11} - \sqrt{4x+3}}{x-2} \dots ??? \dots \frac{-2}{\sqrt{11}}$

1.



Braille letter

Limit of a function, group *Beta* β -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -5/2} \frac{8}{(-2x-5)^{11}} \dots ??? \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 4^{5-2x^2} \dots ??? \dots$ converge

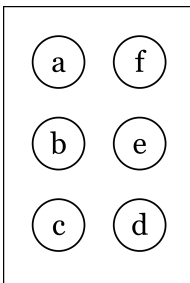
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 9x - 3} - \sqrt{x^2 + x - 6} \right) ??? 8$

(d) $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x-2} \right)^{x/2} \dots ??? \dots e^2$

(e) $\lim_{x \rightarrow -2} \frac{5x^2 + 8x - 4}{2x^3 + 4x^2 - x - 2} \dots ??? \dots -\frac{12}{7}$

(f) $\lim_{x \rightarrow 6/7} \frac{2x - \frac{12}{7}}{\sqrt{3x+1} - \sqrt{7-4x}} \dots ??? \dots \infty$

2.



Braille letter

Limit of a function, group *Beta* β -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -6} \frac{-3}{(x+6)^{11}} \dots ??? \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 7^{5-4x^2} \dots ??? \dots$ diverge

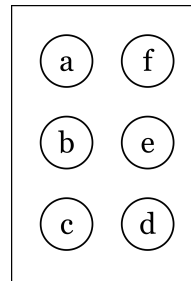
(c) $\lim_{x \rightarrow \infty} \left(4x - \sqrt{16x^2 + 3x - 1} \right) ??? -\frac{3}{4}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{2x-4}{2x-1} \right)^{x/3} \dots ??? \dots \infty$

(e) $\lim_{x \rightarrow -1} \frac{4x^2 + 3x - 1}{-x^3 - x^2 - 8x - 8} \dots ??? \dots \frac{5}{36}$

(f) $\lim_{x \rightarrow 4} \frac{\sqrt{18} - \sqrt{4x+2}}{2x-8} \dots ??? \dots \infty$

3.



Braille letter

Limit of a function, group *Beta* β -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3} \frac{3}{(-3x-9)^{12}} ??? \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 2^{-3-7x^2} \dots ??? \dots$ converge

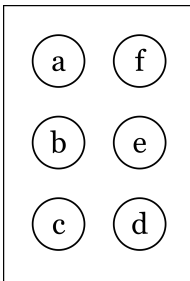
(c) $\lim_{x \rightarrow \infty} \left(7x - \sqrt{49x^2 + 2x - 4} \right) ??? -\frac{1}{7}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{5x-1}{5x+1} \right)^{x/2} \dots ??? \dots \infty$

(e) $\lim_{x \rightarrow -2} \frac{-5x^2 - 16x - 12}{-7x^3 - 14x^2 + 2x + 4} ??? -\frac{2}{13}$

(f) $\lim_{x \rightarrow 7} \frac{\sqrt{17} - \sqrt{3x-4}}{2x-14} \dots ??? \dots \frac{-3}{4\sqrt{17}}$

4.



Braille letter

Limit of a function, group *Gamma* γ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/3} \frac{-5}{(-6x-2)^4}$. ??? . does exist

(b) $\lim_{x \rightarrow \infty} e^{4+3x^2}$??? diverge

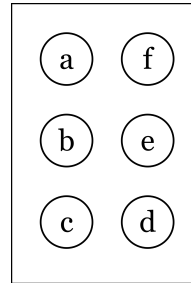
(c) $\lim_{x \rightarrow \infty} (3x - \sqrt{9x^2 + 9x - 3})$. ??? . 3

(d) $\lim_{x \rightarrow \infty} \left(\frac{x-4}{x-3} \right)^{x/3}$??? $e^{-7/3}$

(e) $\lim_{x \rightarrow -3} \frac{4x^2 + 17x + 15}{x^2 - 9}$??? $\frac{7}{6}$

(f) $\lim_{x \rightarrow -3/7} \frac{x + \frac{3}{7}}{\sqrt{3x+2} - \sqrt{-1-4x}}$??? ∞

1.



Braille letter

Limit of a function, group *Gamma* γ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 7/3} \frac{2}{(3x-7)^{15}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{-1-3x^2}$??? diverge

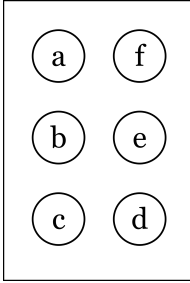
(c) $\lim_{x \rightarrow \infty} (5x - \sqrt{25x^2 - 6x + 2})$??? $\frac{3}{5}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{2x+2}{2x-2} \right)^{x/3}$??? ∞

(e) $\lim_{x \rightarrow -4} \frac{-5x^2 - 25x - 20}{x^2 - 16}$. ??? . $-\frac{15}{8}$

(f) $\lim_{x \rightarrow 8} \frac{\sqrt{58} - \sqrt{7x+2}}{6x-48}$??? 0

2.



Braille letter

Limit of a function, group *Gamma* γ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 2} \frac{1}{(x-2)^{14}}$... ??? ... does exist

(b) $\lim_{x \rightarrow \infty} 4^{3-2x^2}$??? converge

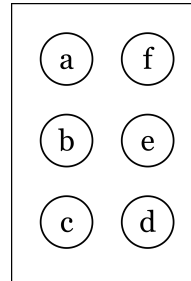
(c) $\lim_{x \rightarrow \infty} (4x - \sqrt{16x^2 - 3x - 1})$??? $\frac{3}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-3x-4}{-3x+3} \right)^{x/9}$??? ∞

(e) $\lim_{x \rightarrow -1} \frac{5x^2 + 3x - 2}{x^2 - 1}$??? $\frac{7}{10}$

(f) $\lim_{x \rightarrow 2} \frac{\sqrt{21} - \sqrt{7x+7}}{3x-6}$.. ??? .. $\frac{-7}{6\sqrt{21}}$

3.



Braille letter

Limit of a function, group *Gamma* γ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/3} \frac{2}{(-9x-3)^{13}}$.. ??? .. does not exist

(b) $\lim_{x \rightarrow \infty} e^{3-7x^2}$??? diverge

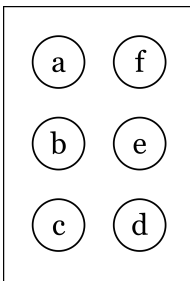
(c) $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x + 3} - \sqrt{x^2 + 4x - 2})$??? 5

(d) $\lim_{x \rightarrow \infty} \left(\frac{-7x-2}{-7x-4} \right)^{x/3}$??? $e^{2/7}$

(e) $\lim_{x \rightarrow -2} \frac{-3x^2 - 14x - 16}{x^2 - 4}$??? $\frac{1}{2}$

(f) $\lim_{x \rightarrow 6/5} \frac{2x - \frac{12}{5}}{\sqrt{3x+1} - \sqrt{7-2x}}$??? 0

4.



Braille letter

Limit of a function, group *Delta* δ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1} \frac{3}{(-6x+6)^6} \dots ??? \dots$ does exist

(b) $\lim_{x \rightarrow \infty} 3^{-4+6x^2} \dots ??? \dots$ diverge

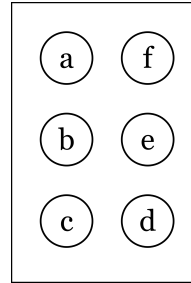
(c) $\lim_{x \rightarrow \infty} (8x - \sqrt{64x^2 - 3x + 3}) \dots -\frac{3}{16}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{3x+8}{3x-9} \right)^{x/8} \dots \infty$

(e) $\lim_{x \rightarrow -3} \frac{-2x^2 - 5x + 3}{-3x^3 - 9x^2 - 3x - 9} \dots -\frac{7}{30}$

(f) $\lim_{x \rightarrow -3/2} \frac{2x+3}{\sqrt{2x+9} - \sqrt{3-2x}} \dots 0$

1.



Braille letter

Limit of a function, group *Delta* δ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -5/3} \frac{4}{(3x+5)^2} \dots ??? \dots$ does exist

(b) $\lim_{x \rightarrow \infty} 3^{2+3x^2} \dots ??? \dots$ converge

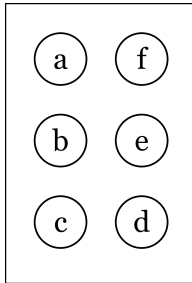
(c) $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 - 7x + 2}) \dots -7$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-9x-1}{-9x+2} \right)^{x/8} \dots e^{-1/72}$

(e) $\lim_{x \rightarrow -2} \frac{-5x^2 - 15x - 10}{-8x^3 - 16x^2 + 4x + 8} \dots 0$

(f) $\lim_{x \rightarrow 2} \frac{\sqrt{13} - \sqrt{7x-1}}{9x-18} \dots \infty$

2.



Braille letter

Limit of a function, group *Delta* δ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 4/7} \frac{4}{(-7x+4)^{12}} \dots ??? \dots$ does exist

(b) $\lim_{x \rightarrow \infty} 4^{1-4x^2} \dots ??? \dots$ diverge

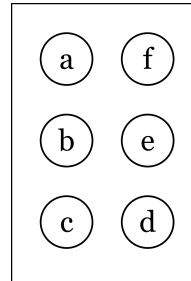
(c) $\lim_{x \rightarrow \infty} (\sqrt{64x^2 - 8x + 1} - \sqrt{64x^2 + 5x - 1}) \dots -\frac{13}{16}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{4x-5}{4x-2} \right)^{x/5} \dots \infty$

(e) $\lim_{x \rightarrow -3} \frac{5x^2 + 14x - 3}{8x^3 + 24x^2 + 2x + 6} \dots -\frac{8}{37}$

(f) $\lim_{x \rightarrow 6} \frac{6 - \sqrt{5x+6}}{2x-12} \dots -\frac{5}{24}$

3.



Braille letter

Limit of a function, group *Delta* δ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 5/2} \frac{3}{(-2x+5)^{13}} \dots ??? \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 3^{6-4x^2} \dots ??? \dots$ diverge

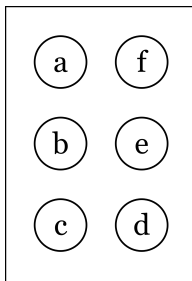
(c) $\lim_{x \rightarrow \infty} (\sqrt{4x^2 - x + 3} - \sqrt{4x^2 + 5x - 1}) \dots 1$

(d) $\lim_{x \rightarrow \infty} \left(\frac{2x-8}{2x-1} \right)^{x/2} \dots e^{-9/4}$

(e) $\lim_{x \rightarrow -4} \frac{-2x^2 - 6x + 8}{-5x^3 - 20x^2 + x + 4} \dots -\frac{10}{79}$

(f) $\lim_{x \rightarrow 3/5} \frac{x - \frac{3}{5}}{\sqrt{2x+1} - \sqrt{4-3x}} \dots \frac{\sqrt{44}}{\sqrt{125}}$

4.



Braille letter

Limit of a function, group *Epsilon* ϵ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1} \frac{-4}{(x-1)^4}$... ??? .. does not exist

(b) $\lim_{x \rightarrow \infty} 2^{-4+3x^2}$??? diverge

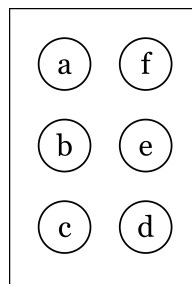
(c) $\lim_{x \rightarrow \infty} (2x - \sqrt{4x^2 + 3x - 1})$??? $-\frac{3}{4}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{x-1}{x+5} \right)^{x/4}$??? ∞

(e) $\lim_{x \rightarrow -1} \frac{2x^2 - 2x - 4}{x^2 - 1}$??? 0

(f) $\lim_{x \rightarrow 5/8} \frac{x - \frac{5}{8}}{\sqrt{4x - 2} - \sqrt{3 - 4x}}$??? $\frac{1}{\sqrt{32}}$

1.



Braille letter

Limit of a function, group *Epsilon* ϵ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1} \frac{-2}{(x+1)^2}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 3^{-3-x^2}$??? converge

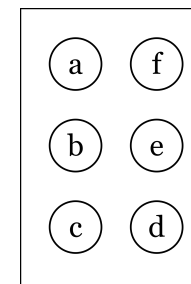
(c) $\lim_{x \rightarrow \infty} (\sqrt{36x^2 + x - 5} - \sqrt{36x^2 - 6x - 6})$??? $-\frac{5}{6}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{2x-1}{2x-5} \right)^{x/8}$??? $e^{-3/8}$

(e) $\lim_{x \rightarrow -4} \frac{-8x^2 - 30x + 8}{x^2 - 16}$??? 0

(f) $\lim_{x \rightarrow -2/3} \frac{4x + \frac{8}{3}}{\sqrt{x+1} - \sqrt{-1-2x}}$??? $\frac{8}{\sqrt{27}}$

2.



Braille letter

Limit of a function, group *Epsilon* ϵ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -4/7} \frac{2}{(7x+4)^7}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{3+4x^2}$??? converge

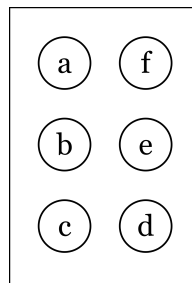
(c) $\lim_{x \rightarrow \infty} (8x - \sqrt{64x^2 + x - 2})$??? $-\frac{1}{16}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x-1}{-2x+1} \right)^{x/6}$??? $e^{1/6}$

(e) $\lim_{x \rightarrow -2} \frac{7x^2 + 6x - 16}{-3x^3 - 6x^2 + 6x + 12}$??? $\frac{11}{3}$

(f) $\lim_{x \rightarrow 5} \frac{\sqrt{7} - \sqrt{x+2}}{5x - 25}$??? ∞

3.



Braille letter

Limit of a function, group *Epsilon* ϵ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/2} \frac{-6}{(4x+2)^{13}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 7^{-2-8x^2}$??? diverge

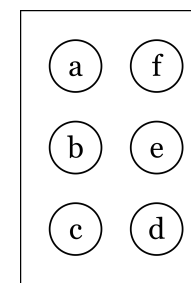
(c) $\lim_{x \rightarrow \infty} (8x - \sqrt{64x^2 - x - 7})$??? $-\frac{1}{16}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-4x-1}{-4x+5} \right)^{x/4}$??? ∞

(e) $\lim_{x \rightarrow -5} \frac{2x^2 + 5x - 25}{-3x^3 - 15x^2 + x + 5}$. ??? . $\frac{15}{74}$

(f) $\lim_{x \rightarrow 2/5} \frac{3x - \frac{6}{5}}{\sqrt{2x+1} - \sqrt{3-3x}}$. ??? . ∞

4.



Braille letter

Limit of a function, group *Zeta* ζ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -7/3} \frac{-5}{(-3x-7)^{14}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 2^{4+x^2}$??? converge

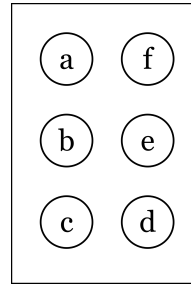
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{16x^2 - x + 3} - \sqrt{16x^2 - 2x + 4} \right)$??? $-\frac{3}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-4x-8}{-4x-3} \right)^{x/3}$??? ∞

(e) $\lim_{x \rightarrow -2} \frac{-4x^2 - 13x - 10}{x^2 - 4}$??? 0

(f) $\lim_{x \rightarrow 11/2} \frac{4x-22}{\sqrt{x-5} - \sqrt{6-x}}$??? $\sqrt{8}$

1.



Braille letter

Limit of a function, group *Zeta* ζ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -7/3} \frac{-2}{(3x+7)^{15}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 5^{-1-x^2}$??? converge

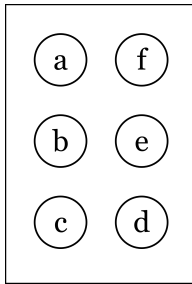
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{4x^2 + 2x - 7} - \sqrt{4x^2 - 3x - 3} \right)$??? $-\frac{1}{4}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-x-7}{-x-2} \right)^{x/2}$??? ∞

(e) $\lim_{x \rightarrow -4} \frac{-6x^2 - 29x - 20}{-4x^3 - 16x^2 - 6x - 24}$??? $\frac{19}{420}$

(f) $\lim_{x \rightarrow 4} \frac{4 - \sqrt{5x-4}}{3x-12}$??? $-\frac{5}{24}$

2.



Braille letter

Limit of a function, group *Zeta* ζ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3/4} \frac{6}{(-8x-6)^6}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{6+3x^2}$??? diverge

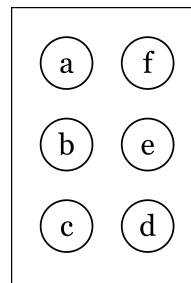
(c) $\lim_{x \rightarrow \infty} \left(2x - \sqrt{4x^2 - 4x - 3} \right)$. ??? . 1

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x-6}{-2x+3} \right)^{x/3}$??? $e^{1/2}$

(e) $\lim_{x \rightarrow -3} \frac{-8x^2 - 28x - 12}{x^2 - 9}$.. ??? .. $-\frac{10}{3}$

(f) $\lim_{x \rightarrow 2} \frac{x-2}{\sqrt{3x-3} - \sqrt{1+x}}$.. ??? .. $\sqrt{3}$

3.



Braille letter

Limit of a function, group *Zeta* ζ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 2/3} \frac{1}{(-6x+4)^8}$??? does exist

(b) $\lim_{x \rightarrow \infty} 4^{4-6x^2}$??? diverge

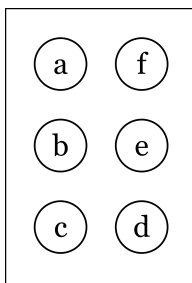
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{9x^2 + 2x - 3} - \sqrt{9x^2 + 4x - 2} \right)$??? $-\frac{1}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{2x+6}{2x-5} \right)^{x/7}$??? $e^{11/14}$

(e) $\lim_{x \rightarrow -3} \frac{4x^2 + 17x + 15}{x^2 - 9}$??? $\frac{7}{6}$

(f) $\lim_{x \rightarrow 6} \frac{\sqrt{10} - \sqrt{2x-2}}{5x-30}$??? $\frac{-1}{5\sqrt{10}}$

4.



Braille letter

Limit of a function, group *Eta* η -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -2} \frac{2}{(x+2)^{10}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 7^{3-x^2}$??? diverge

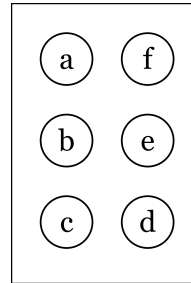
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{25x^2 - 8x + 1} - \sqrt{25x^2 - 4x + 6} \right)$??? $-\frac{4}{5}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{5x-7}{5x+6} \right)^{x/7}$??? ∞

(e) $\lim_{x \rightarrow -4} \frac{4x^2 + 19x + 12}{3x^3 + 12x^2 - 4x - 16}$??? $-\frac{13}{44}$

(f) $\lim_{x \rightarrow 8} \frac{\sqrt{32} - \sqrt{3x+8}}{2x-16}$??? $\frac{-3}{4\sqrt{32}}$

1.



Braille letter

Limit of a function, group *Eta* η -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1} \frac{1}{(-x+1)^{14}}$.. ??? .. does exist

(b) $\lim_{x \rightarrow \infty} 4^{-2-3x^2}$??? diverge

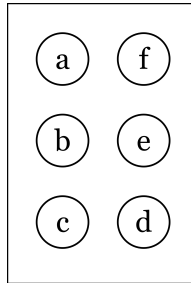
(c) $\lim_{x \rightarrow \infty} \left(3x - \sqrt{9x^2 + 5x + 7} \right)$??? $-\frac{5}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x+4}{-5x+1} \right)^{x/3}$... ??? ... $e^{-1/3}$

(e) $\lim_{x \rightarrow -6} \frac{-2x^2 - 5x + 42}{-2x^3 - 12x^2 - 4x - 24}$??? 0

(f) $\lim_{x \rightarrow -2} \frac{\sqrt{2} - \sqrt{x+4}}{3x+6}$??? 0

2.



Braille letter

Limit of a function, group *Eta* η -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1/4} \frac{-3}{(4x-1)^4}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 3^{8-5x^2}$??? converge

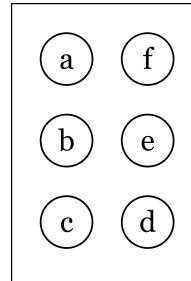
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{25x^2 + 9x + 9} - \sqrt{25x^2 + 3x - 8} \right)$??? $\frac{3}{5}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{8x-8}{8x+2} \right)^{x/3}$??? $e^{-1/4}$

(e) $\lim_{x \rightarrow -2} \frac{6x^2 + 9x - 6}{-2x^3 - 4x^2 + 9x + 18}$??? -15

(f) $\lim_{x \rightarrow 2} \frac{\sqrt{5} - \sqrt{5x-5}}{9x-18}$??? $\frac{-5}{18\sqrt{5}}$

3.



Braille letter

Limit of a function, group *Eta* η -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 7/6} \frac{3}{(6x-7)^9}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{-3-7x^2}$??? diverge

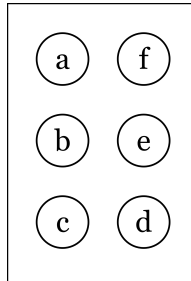
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{64x^2 - 3x + 2} - \sqrt{64x^2 - 7x + 6} \right)$??? $-\frac{5}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-6x+1}{-6x-1} \right)^{x/9}$??? ∞

(e) $\lim_{x \rightarrow -5} \frac{-6x^2 - 36x - 30}{x^2 - 25}$??? $-\frac{12}{5}$

(f) $\lim_{x \rightarrow 3} \frac{\sqrt{6} - \sqrt{3x-3}}{4x-12}$??? 0

4.



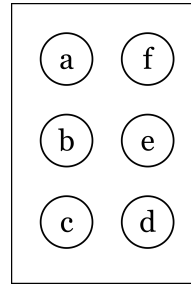
Braille letter

Limit of a function, group *Theta* θ -i

- (a) Determine whether the limit of the function exists. (d) Evaluate the limit of the function.
 (b) Determine whether the limit of the function converges or diverges. (e) Evaluate the limit of the function.
 (c) Evaluate the limit of the function. (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow -1/2} \frac{6}{(-2x-1)^7}$??? does not exist
 (b) $\lim_{x \rightarrow \infty} 6^{-5-5x^2}$??? diverge
 (c) $\lim_{x \rightarrow \infty} \left(\sqrt{36x^2 + 4x - 8} - \sqrt{36x^2 - x - 4} \right)$??? $\frac{5}{12}$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{-x-1}{-x+3} \right)^{x/2}$??? ∞
 (e) $\lim_{x \rightarrow -3} \frac{2x^2 + 8x + 6}{x^2 - 9}$??? $\frac{1}{3}$
 (f) $\lim_{x \rightarrow 1} \frac{\sqrt{5} - \sqrt{x+4}}{2x-2}$??? $\frac{-1}{4\sqrt{5}}$

1.



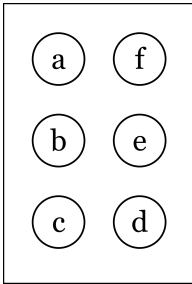
Braille letter

Limit of a function, group *Theta* θ -ii

- (a) Determine whether the limit of the function exists. (d) Evaluate the limit of the function.
 (b) Determine whether the limit of the function converges or diverges. (e) Evaluate the limit of the function.
 (c) Evaluate the limit of the function. (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow 1/2} \frac{-3}{(-4x+2)^3}$. ??? . does exist
 (b) $\lim_{x \rightarrow \infty} 3^{-3+2x^2}$??? diverge
 (c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 - 2x + 3} \right)$. ??? . 2
 (d) $\lim_{x \rightarrow \infty} \left(\frac{-2x+3}{-2x+8} \right)^{x/6}$.. ??? .. $e^{-11/12}$
 (e) $\lim_{x \rightarrow -4} \frac{-4x^2 - 19x - 12}{-3x^3 - 12x^2 + x + 4}$??? $\frac{13}{188}$
 (f) $\lim_{x \rightarrow 8} \frac{\sqrt{11} - \sqrt{x+3}}{5x-40}$.. ??? .. $\frac{-1}{10\sqrt{11}}$

2.



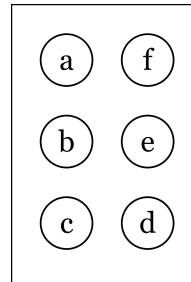
Braille letter

Limit of a function, group *Theta* θ -iii

- (a) Determine whether the limit of the function exists. (d) Evaluate the limit of the function.
 (b) Determine whether the limit of the function converges or diverges. (e) Evaluate the limit of the function.
 (c) Evaluate the limit of the function. (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow 1/2} \frac{-2}{(-6x+3)^2}$.. ??? .. does exist
 (b) $\lim_{x \rightarrow \infty} 7^{-6+4x^2}$??? converge
 (c) $\lim_{x \rightarrow \infty} \left(8x - \sqrt{64x^2 - 3x - 4} \right)$??? $\frac{3}{16}$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{-3x-7}{-3x+4} \right)^{x/2}$??? ∞
 (e) $\lim_{x \rightarrow -2} \frac{5x^2 + x - 18}{x^2 - 4}$??? $\frac{19}{4}$
 (f) $\lim_{x \rightarrow -2/3} \frac{2x + \frac{4}{3}}{\sqrt{4x+5} - \sqrt{3+x}}$??? $\frac{\sqrt{112}}{\sqrt{27}}$

3.



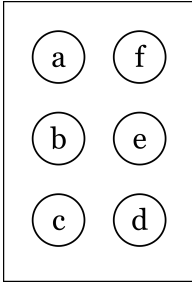
Braille letter

Limit of a function, group *Theta* θ -iv

- (a) Determine whether the limit of the function exists. (d) Evaluate the limit of the function.
 (b) Determine whether the limit of the function converges or diverges. (e) Evaluate the limit of the function.
 (c) Evaluate the limit of the function. (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow -1/3} \frac{6}{(-9x-3)^7}$... ??? ... does not exist
 (b) $\lim_{x \rightarrow \infty} 2^{-6+x^2}$??? converge
 (c) $\lim_{x \rightarrow \infty} \left(\sqrt{4x^2 + x - 2} - \sqrt{4x^2 - x + 5} \right)$??? 0
 (d) $\lim_{x \rightarrow \infty} \left(\frac{9x-3}{9x-1} \right)^{x/4}$??? ∞
 (e) $\lim_{x \rightarrow -3} \frac{-7x^2 - 17x + 12}{2x^3 + 6x^2 - 8x - 24}$??? $\frac{5}{2}$
 (f) $\lim_{x \rightarrow 4} \frac{\sqrt{5} - \sqrt{3x-7}}{3x-12}$??? $\frac{-1}{2\sqrt{5}}$

4.



Braille letter

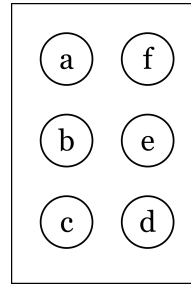
Limit of a function, group *Iota* ι -i

- (a) Determine whether the limit of the function exists.
 (b) Determine whether the limit of the function converges or diverges.
 (c) Evaluate the limit of the function.

- (d) Evaluate the limit of the function.
 (e) Evaluate the limit of the function.
 (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow -7/3} \frac{2}{(3x+7)^4}$. ??? . does exist
 (b) $\lim_{x \rightarrow \infty} e^{-6-x^2}$??? converge
 (c) $\lim_{x \rightarrow \infty} \left(5x - \sqrt{25x^2 + x - 5} \right)$??? $\frac{1}{5}$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{6x-6}{6x-2} \right)^{x/7}$??? ∞
 (e) $\lim_{x \rightarrow -6} \frac{7x^2 + 38x - 24}{x^2 - 36}$.. ??? .. $\frac{23}{42}$
 (f) $\lim_{x \rightarrow 6} \frac{2x-12}{\sqrt{2x-5} - \sqrt{1+x}}$. ??? . ∞

1.



Braille letter

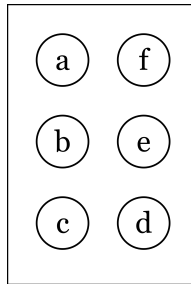
Limit of a function, group *Iota* ι -ii

- (a) Determine whether the limit of the function exists.
 (b) Determine whether the limit of the function converges or diverges.
 (c) Evaluate the limit of the function.

- (d) Evaluate the limit of the function.
 (e) Evaluate the limit of the function.
 (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow 5/3} \frac{-5}{(3x-5)^9}$. ??? . does not exist
 (b) $\lim_{x \rightarrow \infty} 6^{-2-6x^2}$??? diverge
 (c) $\lim_{x \rightarrow \infty} \left(8x - \sqrt{64x^2 - 2x - 8} \right)$??? $-\frac{1}{4}$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{-2x-1}{-2x+1} \right)^{x/4}$??? 1
 (e) $\lim_{x \rightarrow -2} \frac{8x^2 + 17x + 2}{-3x^3 - 6x^2 + 2x + 4}$. ??? . 0
 (f) $\lim_{x \rightarrow 3} \frac{\sqrt{7} - \sqrt{3x-2}}{x-3}$??? 0

2.



Braille letter

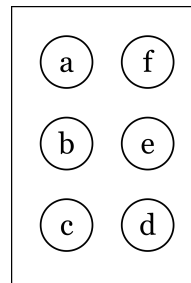
Limit of a function, group *Iota* ι -iii

- (a) Determine whether the limit of the function exists.
 (b) Determine whether the limit of the function converges or diverges.
 (c) Evaluate the limit of the function.

- (d) Evaluate the limit of the function.
 (e) Evaluate the limit of the function.
 (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow 3/2} \frac{-6}{(2x-3)^{12}}$??? does exist
 (b) $\lim_{x \rightarrow \infty} 5^{4+6x^2}$??? diverge
 (c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 5x - 3} - \sqrt{x^2 - 6x - 3} \right)$??? $\frac{1}{2}$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{2x-8}{2x-6} \right)^{x/4}$??? $e^{-7/4}$
 (e) $\lim_{x \rightarrow -4} \frac{7x^2 + 34x + 24}{5x^3 + 20x^2 + 3x + 12}$??? 0
 (f) $\lim_{x \rightarrow 7/3} \frac{x - \frac{7}{3}}{\sqrt{x+1} - \sqrt{8-2x}}$??? 0

3.



Braille letter

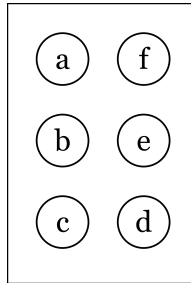
Limit of a function, group *Iota* ι -iv

- (a) Determine whether the limit of the function exists.
 (b) Determine whether the limit of the function converges or diverges.
 (c) Evaluate the limit of the function.

- (d) Evaluate the limit of the function.
 (e) Evaluate the limit of the function.
 (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow -4/3} \frac{-1}{(-3x-4)^{13}}$??? does not exist
 (b) $\lim_{x \rightarrow \infty} 2^{-7+7x^2}$??? diverge
 (c) $\lim_{x \rightarrow \infty} \left(6x - \sqrt{36x^2 + 4x - 3} \right)$??? $-\frac{1}{3}$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{5x-7}{5x+5} \right)^{x/5}$??? ∞
 (e) $\lim_{x \rightarrow -4} \frac{5x^2 + 16x - 16}{x^2 - 16}$??? 0
 (f) $\lim_{x \rightarrow 5} \frac{\sqrt{19} - \sqrt{3x+4}}{2x-10}$??? 0

4.



Braille letter

Limit of a function, group *Kappa* κ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1/5} \frac{1}{(-5x+1)^{12}}$. ??? . does exist

(b) $\lim_{x \rightarrow \infty} 4^{-2+x^2}$??? converge

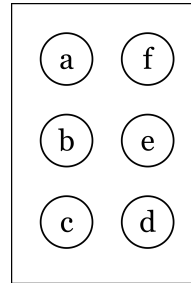
(c) $\lim_{x \rightarrow \infty} \left(5x - \sqrt{25x^2 - 4x + 3}\right)$??? $\frac{2}{5}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{6x-2}{6x-4}\right)^{x/4}$??? $e^{-1/4}$

(e) $\lim_{x \rightarrow -3} \frac{8x^2 + 18x - 18}{x^2 - 9}$... ??? ... 5

(f) $\lim_{x \rightarrow 6/5} \frac{x - \frac{6}{5}}{\sqrt{2x-2} - \sqrt{4-3x}}$??? $\frac{\sqrt{8}}{\sqrt{125}}$

1.



Braille letter

Limit of a function, group *Kappa* κ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 3/2} \frac{-8}{(2x-3)^7}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 6^{6-4x^2}$??? diverge

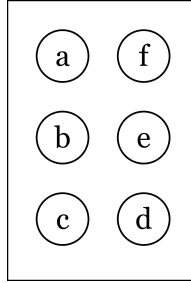
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 - x - 2}\right)$??? $-\frac{1}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-7x+4}{-7x+3}\right)^{x/4}$... ??? ... ∞

(e) $\lim_{x \rightarrow -2} \frac{-3x^2 - 8x - 4}{x^2 - 4}$.. ??? .. -1

(f) $\lim_{x \rightarrow 2} \frac{\sqrt{3} - \sqrt{3x-3}}{2x-4}$... ??? ... ∞

2.



Braille letter

Limit of a function, group *Kappa* κ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3/7} \frac{4}{(7x+3)^6}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 3^{-5+2x^2}$??? diverge

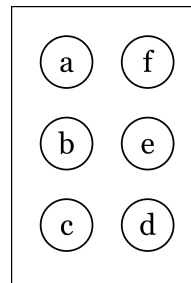
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + 5x - 8}\right)$??? -5

(d) $\lim_{x \rightarrow \infty} \left(\frac{-4x-2}{-4x-1}\right)^{x/2}$... ??? ... $e^{1/8}$

(e) $\lim_{x \rightarrow -1} \frac{-2x^2 + x + 3}{-x^3 - x^2 - 4x - 4}$. ??? . -1

(f) $\lim_{x \rightarrow 1/2} \frac{x - \frac{1}{2}}{\sqrt{3x-1} - \sqrt{2-3x}}$??? $\frac{1}{\sqrt{18}}$

3.



Braille letter

Limit of a function, group *Kappa* κ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1/6} \frac{2}{(-6x+1)^{14}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 7^{5-4x^2}$??? converge

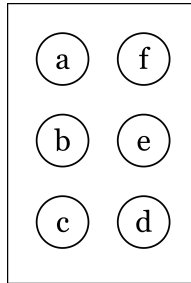
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{49x^2 - 5x - 7} - \sqrt{49x^2 + 7x + 5}\right)$??? $-\frac{6}{7}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{3x+4}{3x+1}\right)^{x/3}$??? $e^{5/9}$

(e) $\lim_{x \rightarrow -4} \frac{-4x^2 - 19x - 12}{x^2 - 16}$??? 0

(f) $\lim_{x \rightarrow -1} \frac{1 - \sqrt{3x+4}}{x+1}$??? $-\frac{3}{2}$

4.



Braille letter

Limit of a function, group *Lambda* λ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/5} \frac{-1}{(5x+1)^4}$??? does exist

(b) $\lim_{x \rightarrow \infty} 4^{-4+4x^2}$... ??? ... diverge

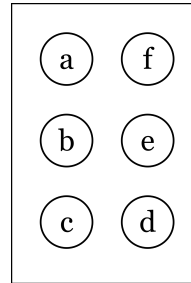
(c) $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + x - 4})$??? -1

(d) $\lim_{x \rightarrow \infty} \left(\frac{-7x+5}{-7x-2} \right)^{x/6}$.. ??? .. ∞

(e) $\lim_{x \rightarrow -1} \frac{6x^2 + 5x - 1}{x^2 - 1}$... ??? ... $\frac{7}{2}$

(f) $\lim_{x \rightarrow 4} \frac{\sqrt{33} - \sqrt{6x+9}}{7x-28}$.. ??? .. ∞

1.



Braille letter

Limit of a function, group *Lambda* λ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/5} \frac{2}{(5x+1)^{11}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 3^{-4-x^2}$??? diverge

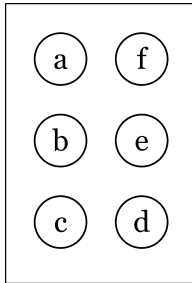
(c) $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 5x - 5})$. ??? . $-\frac{5}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{x-1}{x-4} \right)^{x/6}$??? $e^{-5/6}$

(e) $\lim_{x \rightarrow 5} \frac{7x^2 + 30x - 25}{x^2 - 25}$??? 4

(f) $\lim_{x \rightarrow 7} \frac{\sqrt{11} - \sqrt{2x-3}}{6x-42}$??? 0

2.



Braille letter

Limit of a function, group *Lambda* λ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1} \frac{2}{(2x+2)^6}$... ??? ... does exist

(b) $\lim_{x \rightarrow \infty} 7^{-1+6x^2}$??? converge

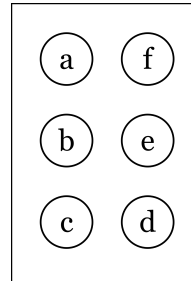
(c) $\lim_{x \rightarrow \infty} (3x - \sqrt{9x^2 + 4x + 2})$??? $-\frac{2}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x+3}{-5x-2} \right)^{x/2}$... ??? ... $e^{-1/10}$

(e) $\lim_{x \rightarrow -7} \frac{7x^2 + 51x + 14}{x^2 - 49}$... ??? ... $\frac{47}{98}$

(f) $\lim_{x \rightarrow 7} \frac{\sqrt{59} - \sqrt{9x-4}}{2x-14}$.. ??? .. $\frac{-9}{4\sqrt{59}}$

3.



Braille letter

Limit of a function, group *Lambda* λ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1/4} \frac{4}{(-4x+1)^4}$??? does exist

(b) $\lim_{x \rightarrow \infty} 3^{5-9x^2}$??? diverge

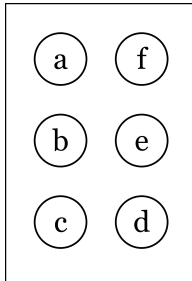
(c) $\lim_{x \rightarrow \infty} (\sqrt{9x^2 + 3x + 3} - \sqrt{9x^2 + 2x + 5})$??? $\frac{1}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-3x-2}{-3x+4} \right)^{x/3}$??? ∞

(e) $\lim_{x \rightarrow -1} \frac{-8x^2 - 9x - 1}{x^2 - 1}$??? $-\frac{7}{2}$

(f) $\lim_{x \rightarrow -1} \frac{1 - \sqrt{7x+8}}{4x+4}$??? ∞

4.



Braille letter

Limit of a function, group $Mu \mu$ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -8/5} \frac{-2}{(5x+8)^{11}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 4^{-2-x^2}$??? converge

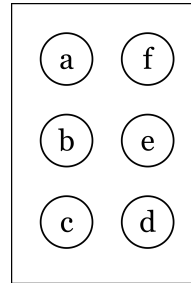
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 2x - 3} - \sqrt{x^2 - x + 4} \right)$??? $-\frac{3}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-4x-4}{-4x+9} \right)^{x/6}$??? $e^{13/24}$

(e) $\lim_{x \rightarrow -3} \frac{-4x^2 - 21x - 27}{x^2 - 9}$??? $-\frac{1}{2}$

(f) $\lim_{x \rightarrow 4/3} \frac{x - \frac{4}{3}}{\sqrt{2x+1} - \sqrt{9-4x}}$??? $\frac{\sqrt{11}}{\sqrt{27}}$

1.



Braille letter

Limit of a function, group $Mu \mu$ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -6} \frac{3}{(x+6)^2}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{-7+3x^2}$??? diverge

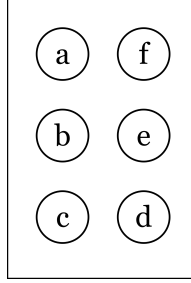
(c) $\lim_{x \rightarrow \infty} \left(8x - \sqrt{64x^2 + x + 5} \right)$??? $\frac{1}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-x-1}{-x+1} \right)^{x/5}$??? ∞

(e) $\lim_{x \rightarrow -1} \frac{-2x^2 - 3x - 1}{x^2 - 1}$... ??? ... $\frac{1}{4}$

(f) $\lim_{x \rightarrow 2} \frac{2x-4}{\sqrt{4x-2} - \sqrt{4+x}}$??? $\frac{\sqrt{32}}{\sqrt{3}}$

2.



Braille letter

Limit of a function, group $Mu \mu$ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 5/4} \frac{8}{(-4x+5)^{11}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 4^{-1-5x^2}$??? converge

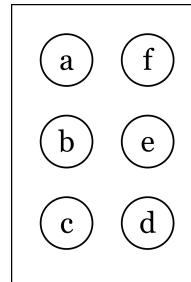
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{64x^2 - x + 1} - \sqrt{64x^2 - 2x + 5} \right)$??? $\frac{1}{16}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{2x+2}{2x+6} \right)^{x/2}$??? ∞

(e) $\lim_{x \rightarrow -2} \frac{2x^2 + 10x + 12}{-x^3 - 2x^2 - 8x - 16}$??? 0

(f) $\lim_{x \rightarrow 8/3} \frac{2x - \frac{16}{3}}{\sqrt{2x-3} - \sqrt{5-x}}$??? $\frac{\sqrt{112}}{\sqrt{27}}$

3.



Braille letter

Limit of a function, group $Mu \mu$ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1/3} \frac{-7}{(9x-3)^2}$.. ??? .. does exist

(b) $\lim_{x \rightarrow \infty} e^{1+x^2}$??? diverge

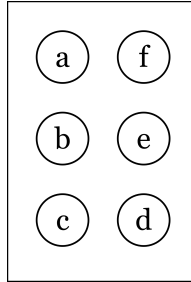
(c) $\lim_{x \rightarrow \infty} \left(3x - \sqrt{9x^2 - 8x + 1} \right)$??? $\frac{8}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{3x+4}{3x+6} \right)^{x/2}$??? $e^{5/3}$

(e) $\lim_{x \rightarrow -4} \frac{8x^2 + 29x - 12}{x^2 - 16}$.. ??? .. $\frac{35}{8}$

(f) $\lim_{x \rightarrow 2} \frac{1 - \sqrt{2x-3}}{x-2}$??? ∞

4.



Braille letter

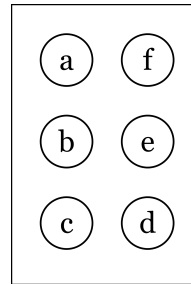
Limit of a function, group $Nu \nu$ -i

- (a) Determine whether the limit of the function exists.
 (b) Determine whether the limit of the function converges or diverges.
 (c) Evaluate the limit of the function.

- (d) Evaluate the limit of the function.
 (e) Evaluate the limit of the function.
 (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow 1/7} \frac{-7}{(7x-1)^2} \dots ??? \dots$ does exist
 (b) $\lim_{x \rightarrow \infty} 4^{6+5x^2} \dots ??? \dots$ converge
 (c) $\lim_{x \rightarrow \infty} (6x - \sqrt{36x^2 + 4x + 1}) \dots ??? \dots -\frac{1}{3}$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{-8x-8}{-8x+5} \right)^{x/2} \dots ??? \dots \infty$
 (e) $\lim_{x \rightarrow -1} \frac{5x^2 - x - 6}{x^2 - 1} \dots ??? \dots \frac{11}{2}$
 (f) $\lim_{x \rightarrow 6} \frac{\sqrt{10} - \sqrt{x+4}}{7x-42} \dots ??? \dots \frac{-1}{14\sqrt{10}}$

1.



Braille letter

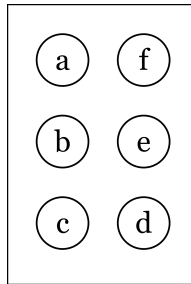
Limit of a function, group $Nu \nu$ -ii

- (a) Determine whether the limit of the function exists.
 (b) Determine whether the limit of the function converges or diverges.
 (c) Evaluate the limit of the function.

- (d) Evaluate the limit of the function.
 (e) Evaluate the limit of the function.
 (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow -7/3} \frac{-5}{(3x+7)^{10}} \dots ??? \dots$ does exist
 (b) $\lim_{x \rightarrow \infty} 2^{4-4x^2} \dots ??? \dots$ diverge
 (c) $\lim_{x \rightarrow \infty} (\sqrt{16x^2 + 3x - 2} - \sqrt{16x^2 + 5x - 4}) \dots ??? \dots 2$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{2x+2}{2x-7} \right)^{x/3} \dots ??? \dots \infty$
 (e) $\lim_{x \rightarrow -5} \frac{4x^2 + 16x - 20}{4x^3 + 20x^2 - x - 5} \dots ??? \dots 0$
 (f) $\lim_{x \rightarrow 4} \frac{\sqrt{28} - \sqrt{6x+4}}{2x-8} \dots ??? \dots 0$

2.



Braille letter

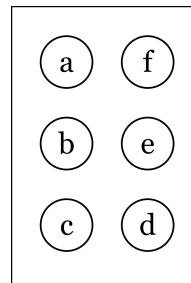
Limit of a function, group $Nu \nu$ -iii

- (a) Determine whether the limit of the function exists.
 (b) Determine whether the limit of the function converges or diverges.
 (c) Evaluate the limit of the function.

- (d) Evaluate the limit of the function.
 (e) Evaluate the limit of the function.
 (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow -2} \frac{4}{(x+2)^{13}} \dots ??? \dots$ does not exist
 (b) $\lim_{x \rightarrow \infty} 3^{-2-x^2} \dots ??? \dots$ diverge
 (c) $\lim_{x \rightarrow \infty} (\sqrt{4x^2 - 7x - 6} - \sqrt{4x^2 - 3x + 6}) \dots ??? \dots -1$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{5x-4}{5x-2} \right)^{x/4} \dots ??? \dots \infty$
 (e) $\lim_{x \rightarrow -4} \frac{-4x^2 - 21x - 20}{2x^3 + 8x^2 + x + 4} \dots ??? \dots 0$
 (f) $\lim_{x \rightarrow 4/3} \frac{5x - \frac{20}{3}}{\sqrt{4x-3} - \sqrt{1+x}} \dots ??? \dots \frac{\sqrt{700}}{\sqrt{27}}$

3.



Braille letter

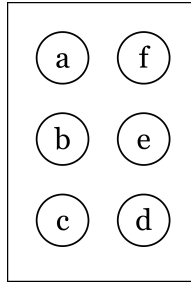
Limit of a function, group $Nu \nu$ -iv

- (a) Determine whether the limit of the function exists.
 (b) Determine whether the limit of the function converges or diverges.
 (c) Evaluate the limit of the function.

- (d) Evaluate the limit of the function.
 (e) Evaluate the limit of the function.
 (f) Evaluate the limit of the function.

- (a) $\lim_{x \rightarrow 1} \frac{-1}{(4x-4)^{10}} \dots ??? \dots$ does exist
 (b) $\lim_{x \rightarrow \infty} 2^{3-2x^2} \dots ??? \dots$ diverge
 (c) $\lim_{x \rightarrow \infty} (\sqrt{49x^2 - 5x - 2} - \sqrt{49x^2 - 2x - 3}) \dots ??? \dots -1$
 (d) $\lim_{x \rightarrow \infty} \left(\frac{3x+4}{3x-4} \right)^{x/4} \dots ??? \dots 1$
 (e) $\lim_{x \rightarrow -5} \frac{-5x^2 - 16x + 45}{-2x^3 - 10x^2 + 5x + 25} \dots ??? \dots -\frac{34}{45}$
 (f) $\lim_{x \rightarrow 2} \frac{4x-8}{\sqrt{2x-3} - \sqrt{-1+x}} \dots ??? \dots 0$

4.



Braille letter

Limit of a function, group $Xi \xi$ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -6/5} \frac{4}{(5x+6)^{12}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 5^{-9+4x^2}$??? diverge

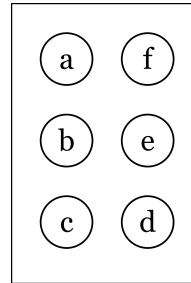
(c) $\lim_{x \rightarrow \infty} \left(8x - \sqrt{64x^2 + 6x - 5}\right)$??? $\frac{3}{4}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x+3}{-2x+1}\right)^{x/2}$??? ∞

(e) $\lim_{x \rightarrow -5} \frac{-8x^2 - 42x - 10}{-x^3 - 5x^2 - 2x - 10}$. ??? . 0

(f) $\lim_{x \rightarrow 1} \frac{\sqrt{6} - \sqrt{x+5}}{x-1}$??? $\frac{-1}{2\sqrt{6}}$

1.



Braille letter

Limit of a function, group $Xi \xi$ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -2/3} \frac{-3}{(6x+4)^{12}}$. ??? . does exist

(b) $\lim_{x \rightarrow \infty} 7^{1-9x^2}$??? diverge

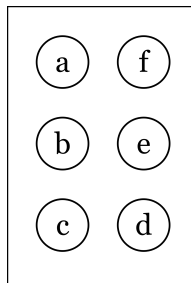
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + 4x - 4}\right)$.. ??? .. 2

(d) $\lim_{x \rightarrow \infty} \left(\frac{4x+2}{4x+7}\right)^{x/2}$??? $e^{9/8}$

(e) $\lim_{x \rightarrow -1} \frac{3x^2 + 10x + 7}{x^2 - 1}$... ??? ... -2

(f) $\lim_{x \rightarrow -3/5} \frac{x + \frac{3}{5}}{\sqrt{x+9} - \sqrt{6-4x}}$??? $\frac{\sqrt{168}}{\sqrt{125}}$

2.



Braille letter

Limit of a function, group $Xi \xi$ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 7/2} \frac{-6}{(2x-7)^{10}}$. ??? . does exist

(b) $\lim_{x \rightarrow \infty} e^{-6+8x^2}$??? converge

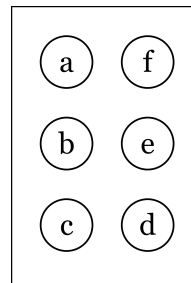
(c) $\lim_{x \rightarrow \infty} \left(3x - \sqrt{9x^2 + 7x - 4}\right)$??? $\frac{7}{6}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-3x+1}{-3x-4}\right)^{x/6}$... ??? ... ∞

(e) $\lim_{x \rightarrow -1} \frac{2x^2 - 4x - 6}{x^3 + x^2 + 2x + 2}$. ??? . $-\frac{8}{3}$

(f) $\lim_{x \rightarrow -1/6} \frac{4x + \frac{2}{3}}{\sqrt{4x+2} - \sqrt{1-2x}}$??? ∞

3.



Braille letter

Limit of a function, group $Xi \xi$ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -9/2} \frac{-4}{(2x+9)^{13}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 3^{4-4x^2}$??? diverge

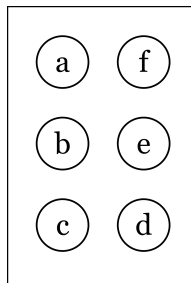
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 - 2x + 6}\right)$. ??? . -1

(d) $\lim_{x \rightarrow \infty} \left(\frac{4x+1}{4x+3}\right)^{x/2}$??? ∞

(e) $\lim_{x \rightarrow -1} \frac{-6x^2 + 3x + 9}{x^2 - 1}$??? $\frac{5}{4}$

(f) $\lim_{x \rightarrow 1/2} \frac{x - \frac{1}{2}}{\sqrt{x+1} - \sqrt{3-3x}}$.. ??? .. ∞

4.



Braille letter

Limit of a function, group *Omicron o* -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -2} \frac{-2}{(-3x-6)^5} \dots \dots \text{???} \dots \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 3^{-1+3x^2} \dots \dots \text{???} \dots \dots$ converge

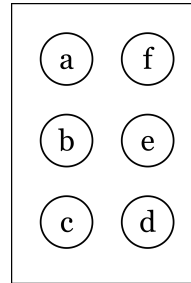
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{4x^2 - 5x + 3} - \sqrt{4x^2 + 9x - 8} \right) \text{???} \quad 2$

(d) $\lim_{x \rightarrow \infty} \left(\frac{3x+5}{3x+8} \right)^{x/2} \dots \dots \text{???} \dots \dots \infty$

(e) $\lim_{x \rightarrow -4} \frac{4x^2 + 25x + 36}{-3x^3 - 12x^2 - 2x - 8} \dots \text{???} \dots \frac{7}{200}$

(f) $\lim_{x \rightarrow 3} \frac{\sqrt{5} - \sqrt{x+2}}{7x-21} \dots \dots \text{???} \dots \dots \frac{-1}{14\sqrt{5}}$

1.



Braille letter

Limit of a function, group *Omicron o* -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 4/5} \frac{3}{(-5x+4)^{11}} \dots \dots \text{???} \dots \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 7^{5-2x^2} \dots \dots \text{???} \dots \dots$ diverge

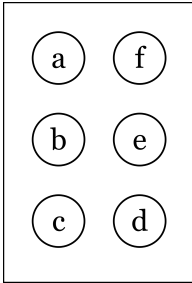
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{4x^2 + 3x + 4} - \sqrt{4x^2 + 4x - 2} \right) \text{???} \quad \frac{7}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x+2}{-2x+6} \right)^{x/4} \dots \dots \text{???} \dots \dots e^{-1}$

(e) $\lim_{x \rightarrow -4} \frac{3x^2 + 15x + 12}{4x^3 + 16x^2 - 4x - 16} \dots \dots \text{???} \dots \dots 0$

(f) $\lim_{x \rightarrow -2/3} \frac{3x+2}{\sqrt{x+3} - \sqrt{1-2x}} \dots \dots \text{???} \dots \dots \infty$

2.



Braille letter

Limit of a function, group *Omicron o* -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3} \frac{-1}{(-x-3)^5} \dots \dots \text{???} \dots \dots$ does exist

(b) $\lim_{x \rightarrow \infty} 3^{9+x^2} \dots \dots \text{???} \dots \dots$ diverge

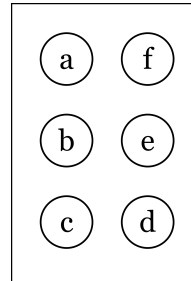
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{64x^2 - 2x - 6} - \sqrt{64x^2 + 7x + 3} \right) \text{???} \quad -\frac{9}{16}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{2x-8}{2x+2} \right)^{x/5} \dots \dots \text{???} \dots \dots \infty$

(e) $\lim_{x \rightarrow -3} \frac{4x^2 + 8x - 12}{-5x^3 - 15x^2 + 5x + 15} \dots \dots \text{???} \dots \dots \frac{1}{10}$

(f) $\lim_{x \rightarrow 8} \frac{\sqrt{32} - \sqrt{5x-8}}{4x-32} \dots \dots \text{???} \dots \dots \frac{-5}{8\sqrt{32}}$

3.



Braille letter

Limit of a function, group *Omicron o* -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 3/5} \frac{2}{(-5x+3)^3} \dots \dots \text{???} \dots \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 3^{-6+x^2} \dots \dots \text{???} \dots \dots$ converge

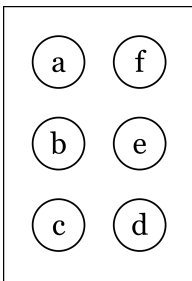
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{25x^2 + 3x + 6} - \sqrt{25x^2 - x - 6} \right) \text{???} \quad \frac{4}{5}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x-3}{-2x+1} \right)^{x/3} \dots \dots \text{???} \dots \dots e^{1/3}$

(e) $\lim_{x \rightarrow -1} \frac{3x^2 - 6x - 9}{x^2 - 1} \dots \dots \text{???} \dots \dots 6$

(f) $\lim_{x \rightarrow -3/5} \frac{2x + \frac{6}{5}}{\sqrt{x+2} - \sqrt{-1-4x}} \dots \dots \text{???} \dots \dots \infty$

4.



Braille letter

Limit of a function, group $Pi \pi$ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/5} \frac{-3}{(-5x-1)^9} \dots \dots \text{???} \dots \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 7^{-8-x^2} \dots \dots \text{???} \dots \dots$ converge

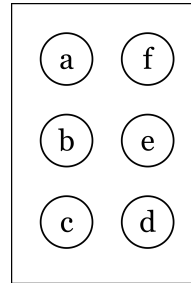
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{64x^2 - 3x + 4} - \sqrt{64x^2 - 4x - 1} \right) \text{???} \frac{1}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x-3}{-5x+3} \right)^{x/7} \dots \dots \text{???} \dots \dots 1$

(e) $\lim_{x \rightarrow -1} \frac{-8x^2 - 14x - 6}{x^2 - 1} \dots \dots \text{???} \dots \dots \frac{1}{8}$

(f) $\lim_{x \rightarrow -6/5} \frac{x + \frac{6}{5}}{\sqrt{4x+5} - \sqrt{-1-x}} \dots \dots \text{???} \dots \dots \frac{2}{\sqrt{125}}$

1.



Braille letter

Limit of a function, group $Pi \pi$ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -2} \frac{1}{(-x-2)^{11}} \text{???} \dots \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 3^{4+x^2} \dots \dots \text{???} \dots \dots$ converge

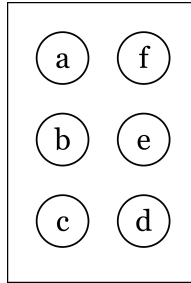
(c) $\lim_{x \rightarrow \infty} \left(6x - \sqrt{36x^2 - 4x + 9} \right) \text{???} \frac{1}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-3x+4}{-3x+3} \right)^{x/9} \dots \dots \text{???} \dots \dots \infty$

(e) $\lim_{x \rightarrow -1} \frac{8x^2 + 13x + 5}{x^2 - 1} \dots \dots \text{???} \dots \dots \frac{3}{2}$

(f) $\lim_{x \rightarrow 1/3} \frac{2x - \frac{2}{3}}{\sqrt{2x+5} - \sqrt{7-4x}} \dots \dots \text{???} \dots \dots \infty$

2.



Braille letter

Limit of a function, group $Pi \pi$ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1} \frac{-3}{(5x+5)^5} \dots \dots \text{???} \dots \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 2^{-3-7x^2} \dots \dots \text{???} \dots \dots$ diverge

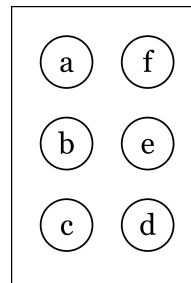
(c) $\lim_{x \rightarrow \infty} \left(2x - \sqrt{4x^2 + x - 9} \right) \text{???} -\frac{1}{4}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x+2}{-5x-5} \right)^{x/3} \dots \dots \text{???} \dots \dots \infty$

(e) $\lim_{x \rightarrow -1} \frac{2x^2 + 7x + 5}{x^2 - 1} \dots \dots \text{???} \dots \dots -\frac{3}{2}$

(f) $\lim_{x \rightarrow -14/5} \frac{x + \frac{14}{5}}{\sqrt{3x+9} - \sqrt{-5-2x}} \text{???} \dots \dots \infty$

3.



Braille letter

Limit of a function, group $Pi \pi$ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3} \frac{1}{(-x-3)^{17}} \dots \dots \text{???} \dots \dots$ does exist

(b) $\lim_{x \rightarrow \infty} 3^{3+7x^2} \dots \dots \text{???} \dots \dots$ diverge

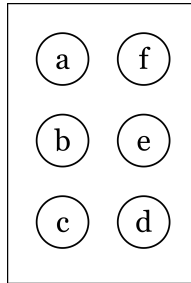
(c) $\lim_{x \rightarrow \infty} \left(3x - \sqrt{9x^2 + 9x - 5} \right) \text{???} -\frac{3}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x-4}{-5x+3} \right)^{x/7} \dots \dots \text{???} \dots \dots \infty$

(e) $\lim_{x \rightarrow -3} \frac{6x^2 + 14x - 12}{x^2 - 9} \dots \dots \text{???} \dots \dots \frac{11}{3}$

(f) $\lim_{x \rightarrow 3} \frac{\sqrt{20} - \sqrt{6x+2}}{3x-9} \dots \dots \text{???} \dots \dots \frac{-1}{\sqrt{20}}$

4.



Braille letter

Limit of a function, group *Rho* ρ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3/4} \frac{-7}{(4x+3)^6}$??? does exist

(b) $\lim_{x \rightarrow \infty} 3^{-3-x^2}$??? diverge

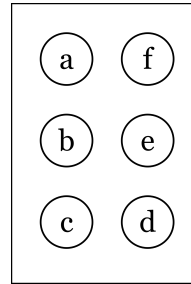
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 - 3x + 1} \right)$??? -1

(d) $\lim_{x \rightarrow \infty} \left(\frac{4x-7}{4x+8} \right)^{x/3}$??? $e^{1/12}$

(e) $\lim_{x \rightarrow -3} \frac{5x^2 + 11x - 12}{2x^3 + 6x^2 + 7x + 21}$??? 0

(f) $\lim_{x \rightarrow 9} \frac{\sqrt{41} - \sqrt{5x-4}}{x-9}$??? 0

1.



Braille letter

Limit of a function, group *Rho* ρ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -4} \frac{1}{(-x-4)^2}$??? does exist

(b) $\lim_{x \rightarrow \infty} 2^{-4-6x^2}$??? diverge

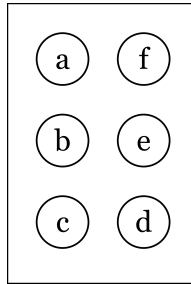
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 2x + 6} - \sqrt{x^2 + 6x + 5} \right)$??? 4

(d) $\lim_{x \rightarrow \infty} \left(\frac{7x-2}{7x+4} \right)^{x/8}$??? ∞

(e) $\lim_{x \rightarrow -4} \frac{6x^2 + 28x + 16}{-2x^3 - 8x^2 + 2x + 8}$??? $\frac{1}{9}$

(f) $\lim_{x \rightarrow 4} \frac{\sqrt{27} - \sqrt{8x-5}}{2x-8}$??? $\frac{-2}{\sqrt{27}}$

2.



Braille letter

Limit of a function, group *Rho* ρ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -6/7} \frac{2}{(7x+6)^8}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{2-2x^2}$??? converge

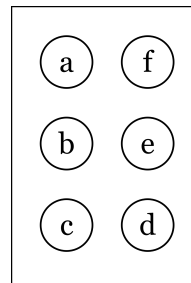
(c) $\lim_{x \rightarrow \infty} \left(3x - \sqrt{9x^2 + x - 7} \right)$. ??? . $\frac{1}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{4x-9}{4x+6} \right)^{x/5}$??? $e^{-3/20}$

(e) $\lim_{x \rightarrow -5} \frac{7x^2 + 33x - 10}{x^3 + 5x^2 - x - 5}$... ??? ... 0

(f) $\lim_{x \rightarrow -1/5} \frac{2x + \frac{2}{5}}{\sqrt{3x+3} - \sqrt{2-2x}}$??? $\frac{\sqrt{192}}{\sqrt{125}}$

3.



Braille letter

Limit of a function, group *Rho* ρ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3} \frac{-7}{(x+3)^{11}}$. ??? . does not exist

(b) $\lim_{x \rightarrow \infty} 5^{3-5x^2}$??? diverge

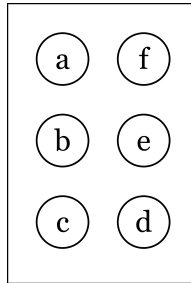
(c) $\lim_{x \rightarrow \infty} \left(6x - \sqrt{36x^2 + 5x + 3} \right)$??? $-\frac{5}{6}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-6x+1}{-6x-4} \right)^{x/4}$??? $e^{1/8}$

(e) $\lim_{x \rightarrow -4} \frac{5x^2 + 22x + 8}{-4x^3 - 16x^2 - 7x - 28}$??? $\frac{18}{71}$

(f) $\lim_{x \rightarrow 1} \frac{\sqrt{11} - \sqrt{8x+3}}{3x-3}$... ??? ... $\frac{-4}{3\sqrt{11}}$

4.



Braille letter

Limit of a function, group *Sigma* σ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -9/7} \frac{1}{(7x+9)^6}$??? does exist

(b) $\lim_{x \rightarrow \infty} 6^{-3-x^2}$??? converge

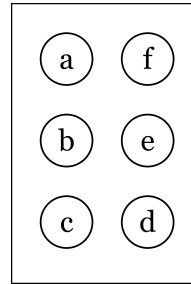
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2+5x-3} - \sqrt{x^2-7x-5} \right)$??? -1

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x-4}{-2x-6} \right)^{x/2}$??? ∞

(e) $\lim_{x \rightarrow -4} \frac{-3x^2-9x+12}{3x^3+12x^2-4x-16}$??? $-\frac{5}{44}$

(f) $\lim_{x \rightarrow 7} \frac{x-7}{\sqrt{4x-4}-\sqrt{3+3x}}$??? ∞

1.



Braille letter

Limit of a function, group *Sigma* σ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/6} \frac{-7}{(-6x-1)^8}$??? does exist

(b) $\lim_{x \rightarrow \infty} 5^{-1+4x^2}$??? converge

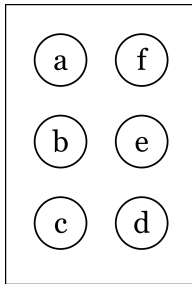
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{25x^2-x-3} - \sqrt{25x^2-6x+1} \right)$??? $\frac{1}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x+5}{-5x+6} \right)^{x/6}$??? ∞

(e) $\lim_{x \rightarrow -1} \frac{2x^2-x-3}{x^2-1}$??? $\frac{5}{2}$

(f) $\lim_{x \rightarrow 4} \frac{\sqrt{12}-\sqrt{2x+4}}{7x-28}$??? ∞

2.



Braille letter

Limit of a function, group *Sigma* σ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/2} \frac{5}{(4x+2)^7}$. ??? . does exist

(b) $\lim_{x \rightarrow \infty} e^{5-6x^2}$??? converge

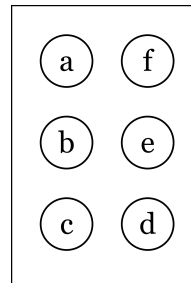
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2+6x-2} \right)$??? -3

(d) $\lim_{x \rightarrow \infty} \left(\frac{4x+2}{4x+9} \right)^{x/3}$... ??? ... $e^{11/12}$

(e) $\lim_{x \rightarrow -1} \frac{-3x^2-5x-2}{8x^3+8x^2-5x-5}$??? $-\frac{1}{9}$

(f) $\lim_{x \rightarrow 9} \frac{\sqrt{43}-\sqrt{5x-2}}{x-9}$. ??? . $\frac{-5}{2\sqrt{43}}$

3.



Braille letter

Limit of a function, group *Sigma* σ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 3/2} \frac{8}{(-2x+3)^2}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{-5+2x^2}$??? diverge

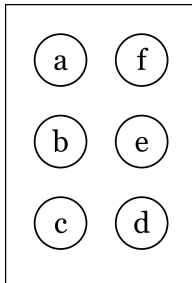
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{4x^2-5x-2} - \sqrt{4x^2+2x-2} \right)$??? $-\frac{7}{4}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{5x-4}{5x+1} \right)^{x/5}$??? $e^{-3/25}$

(e) $\lim_{x \rightarrow -6} \frac{-3x^2-21x-18}{x^3+6x^2-2x-12}$??? 0

(f) $\lim_{x \rightarrow 2/3} \frac{3x-2}{\sqrt{2x+1}-\sqrt{5-4x}}$??? $\frac{\sqrt{7}}{\sqrt{3}}$

4.



Braille letter

Limit of a function, group *Tau* τ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3} \frac{-6}{(-x-3)^{17}} \dots ??? \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 6^{-1+2x^2} \dots ??? \dots$ diverge

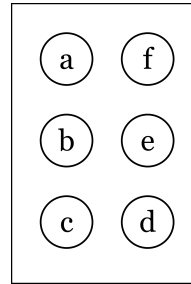
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 5x + 7} - \sqrt{x^2 - 8x - 1} \right) ??? \frac{13}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-x-1}{-x-7} \right)^{x/3} \dots ??? \dots \infty$

(e) $\lim_{x \rightarrow -1} \frac{7x^2 - x - 8}{8x^3 + 8x^2 + x + 1} \dots ??? \dots 0$

(f) $\lim_{x \rightarrow 4} \frac{\sqrt{24} - \sqrt{7x-4}}{2x-8} \dots ??? \dots \frac{-7}{4\sqrt{24}}$

1.



Braille letter

Limit of a function, group *Tau* τ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 2/3} \frac{-7}{(-3x+2)^7} ???$ does not exist

(b) $\lim_{x \rightarrow \infty} e^{-2+8x^2} \dots ??? \dots$ diverge

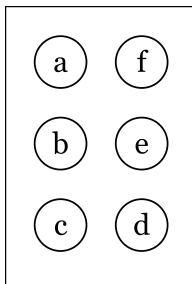
(c) $\lim_{x \rightarrow \infty} \left(9x - \sqrt{81x^2 + 3x - 4} \right) ??? -\frac{1}{6}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x+1}{-2x-4} \right)^{x/8} \dots ??? \dots e^{3/16}$

(e) $\lim_{x \rightarrow -2} \frac{5x^2 + 3x - 14}{-x^3 - 2x^2 - x - 2} \dots ??? \dots 0$

(f) $\lim_{x \rightarrow 4/7} \frac{3x - \frac{12}{7}}{\sqrt{4x-2} - \sqrt{2-3x}} \dots ??? \dots \infty$

2.



Braille letter

Limit of a function, group *Tau* τ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 2/5} \frac{2}{(-5x+2)^2} \dots ??? \dots$ does exist

(b) $\lim_{x \rightarrow \infty} 3^{-2+6x^2} \dots ??? \dots$ converge

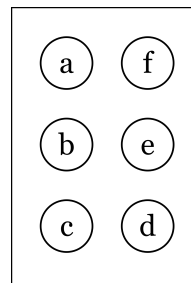
(c) $\lim_{x \rightarrow \infty} \left(8x - \sqrt{64x^2 - 2x - 1} \right) ??? -\frac{1}{4}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{x-4}{x+5} \right)^{x/4} \dots ??? \dots e^{1/4}$

(e) $\lim_{x \rightarrow -3} \frac{4x^2 + 3x - 27}{x^2 - 9} \dots ??? \dots \frac{7}{8}$

(f) $\lim_{x \rightarrow 1} \frac{3 - \sqrt{6x+3}}{8x-8} \dots ??? \dots \infty$

3.



Braille letter

Limit of a function, group *Tau* τ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/2} \frac{-4}{(-6x-3)^3} ???$ does not exist

(b) $\lim_{x \rightarrow \infty} e^{-1-2x^2} \dots ??? \dots$ diverge

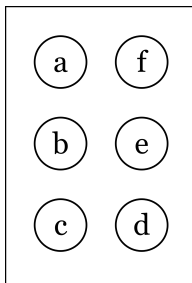
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 - 3x - 2} \right) \dots ??? \dots \frac{3}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-7x-2}{-7x+9} \right)^{x/6} \dots ??? \dots e^{-1/6}$

(e) $\lim_{x \rightarrow -5} \frac{-5x^2 - 24x + 5}{x^2 - 25} \dots ??? \dots -\frac{13}{5}$

(f) $\lim_{x \rightarrow 2} \frac{x-2}{\sqrt{3x-5} - \sqrt{9-4x}} \dots ??? \dots \frac{2}{7}$

4.



Braille letter

Limit of a function, group *Upsilon v* -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 5/2} \frac{-2}{(-2x+5)^7}$??? does exist

(b) $\lim_{x \rightarrow \infty} 3^{-2+9x^2}$??? diverge

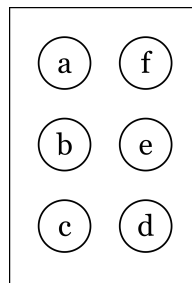
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{4x^2+5x-1} - \sqrt{4x^2+5x+1} \right)$??? 0

(d) $\lim_{x \rightarrow \infty} \left(\frac{-3x+1}{-3x+3} \right)^{x/9}$??? ∞

(e) $\lim_{x \rightarrow -2} \frac{-3x^2-11x-10}{-2x^3-4x^2-4x-8}$??? 0

(f) $\lim_{x \rightarrow 14} \frac{2x-28}{\sqrt{2x-6}-\sqrt{8+x}}$??? $\sqrt{352}$

1.



Braille letter

Limit of a function, group *Upsilon v* -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1} \frac{3}{(-x-1)^{10}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 6^{5+x^2}$??? converge

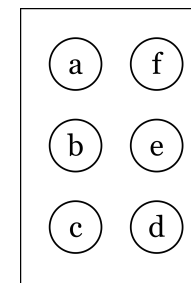
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{49x^2+5x-7} - \sqrt{49x^2+9x-3} \right)$??? $-\frac{4}{7}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-x-4}{-x+2} \right)^{x/3}$??? ∞

(e) $\lim_{x \rightarrow -4} \frac{-5x^2-21x-4}{x^2-16}$??? $\frac{19}{40}$

(f) $\lim_{x \rightarrow 3} \frac{\sqrt{7}-\sqrt{4x-5}}{5x-15}$??? 0

2.



Braille letter

Limit of a function, group *Upsilon v* -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -3/4} \frac{-6}{(4x+3)^7}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{-4-6x^2}$??? converge

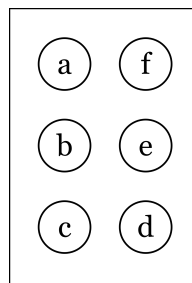
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{16x^2+9x-2} - \sqrt{16x^2+6x-3} \right)$??? $\frac{3}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{6x+3}{6x-4} \right)^{x/4}$??? ∞

(e) $\lim_{x \rightarrow -3} \frac{6x^2+26x+24}{x^3+3x^2-x-3}$??? 0

(f) $\lim_{x \rightarrow 2} \frac{1-\sqrt{3x-5}}{9x-18}$??? 0

3.



Braille letter

Limit of a function, group *Upsilon v* -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -7} \frac{-6}{(x+7)^{17}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 7^{4+x^2}$??? converge

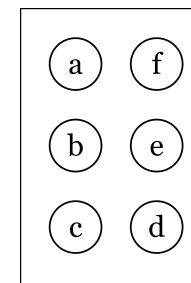
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{81x^2-2x-6} - \sqrt{81x^2-4x+4} \right)$??? $-\frac{2}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x+2} \right)^{x/5}$??? $e^{3/5}$

(e) $\lim_{x \rightarrow -1} \frac{-2x^2+x+3}{x^2-1}$??? $-\frac{5}{2}$

(f) $\lim_{x \rightarrow 9} \frac{\sqrt{77}-\sqrt{9x-4}}{3x-27}$??? ∞

4.



Braille letter

Limit of a function, group *Phi* ϕ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -2} \frac{-2}{(-3x-6)^{10}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 4^{-4-3x^2}$??? diverge

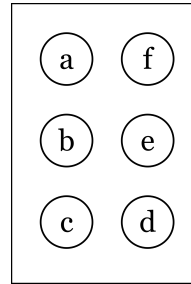
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{9x^2 + x + 3} - \sqrt{9x^2 + 5x + 4} \right)$??? $-\frac{4}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{5x+6}{5x+4} \right)^{x/5}$??? ∞

(e) $\lim_{x \rightarrow -2} \frac{-3x^2 - 13x - 14}{x^2 - 4}$??? 0

(f) $\lim_{x \rightarrow -7/4} \frac{x + \frac{7}{4}}{\sqrt{2x+6} - \sqrt{-1-2x}}$??? ∞

1.



Braille letter

Limit of a function, group *Phi* ϕ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 7/3} \frac{-4}{(-3x+7)^7}$??? does exist

(b) $\lim_{x \rightarrow \infty} 4^{1-9x^2}$??? converge

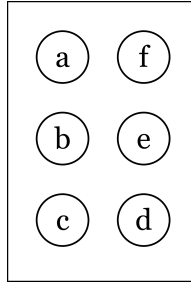
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{25x^2 + 2x + 4} - \sqrt{25x^2 - x - 1} \right)$??? $\frac{3}{10}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{5x-1}{5x+1} \right)^{x/6}$??? ∞

(e) $\lim_{x \rightarrow -1} \frac{-9x^2 - 8x + 1}{-x^3 - x^2 - 2x - 2}$??? $-\frac{10}{3}$

(f) $\lim_{x \rightarrow 5} \frac{\sqrt{21} - \sqrt{4x+1}}{x-5}$??? $\frac{-2}{\sqrt{21}}$

2.



Braille letter

Limit of a function, group *Phi* ϕ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1/8} \frac{9}{(-8x-1)^{13}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 6^{-4+x^2}$??? converge

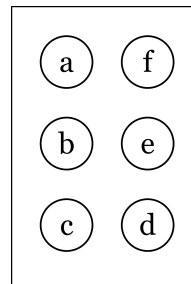
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{36x^2 + 8x - 8} - \sqrt{36x^2 + 2x + 5} \right)$??? $\frac{1}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{3x+2}{3x-2} \right)^{x/4}$??? ∞

(e) $\lim_{x \rightarrow -2} \frac{2x^2 + 9x + 10}{-6x^3 - 12x^2 - 2x - 4}$??? $-\frac{1}{26}$

(f) $\lim_{x \rightarrow -9/5} \frac{x + \frac{9}{5}}{\sqrt{2x+6} - \sqrt{-3-3x}}$??? ∞

3.



Braille letter

Limit of a function, group *Phi* ϕ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 2} \frac{-1}{(x-2)^{12}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 6^{-6-x^2}$??? diverge

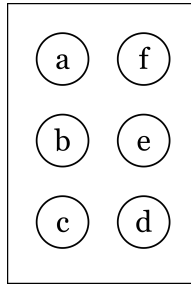
(c) $\lim_{x \rightarrow \infty} \left(6x - \sqrt{36x^2 + x - 4} \right)$??? $-\frac{1}{12}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-4x+3}{-4x+6} \right)^{x/2}$??? $e^{-9/8}$

(e) $\lim_{x \rightarrow -5} \frac{-7x^2 - 30x + 25}{2x^3 + 10x^2 + 3x + 15}$??? $-\frac{40}{371}$

(f) $\lim_{x \rightarrow 2} \frac{\sqrt{13} - \sqrt{4x+5}}{x-2}$??? $\frac{-2}{\sqrt{13}}$

4.



Braille letter

Limit of a function, group *Chi* χ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1/2} \frac{-3}{(4x-2)^{12}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 3^{-3+2x^2}$??? diverge

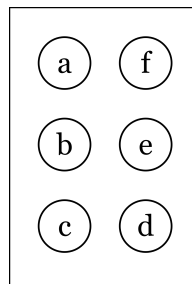
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{36x^2 - 4x + 4} - \sqrt{36x^2 + x - 5} \right)$??? $-\frac{5}{12}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x+2}{-5x-7} \right)^{x/3}$??? $e^{1/3}$

(e) $\lim_{x \rightarrow -1} \frac{7x^2 - x - 8}{7x^3 + 7x^2 + 2x + 2}$??? $-\frac{5}{21}$

(f) $\lim_{x \rightarrow -3/5} \frac{x + \frac{3}{5}}{\sqrt{4x+5} - \sqrt{2-x}}$??? $\frac{\sqrt{52}}{\sqrt{125}}$

1.



Braille letter

Limit of a function, group *Chi* χ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1} \frac{5}{(2x+2)^{11}}$. ??? . does exist

(b) $\lim_{x \rightarrow \infty} 3^{4+2x^2}$??? diverge

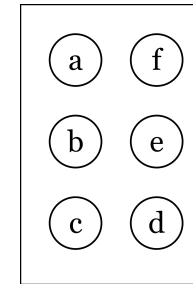
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + 6x - 2} \right)$??? -3

(d) $\lim_{x \rightarrow \infty} \left(\frac{4x+1}{4x+3} \right)^{x/8}$... ??? ... $e^{1/8}$

(e) $\lim_{x \rightarrow -1} \frac{5x^2 - x - 6}{-4x^3 - 4x^2 + x + 1}$??? $\frac{11}{3}$

(f) $\lim_{x \rightarrow -2} \frac{\sqrt{5} - \sqrt{x+7}}{x+2}$.. ??? .. $\frac{-1}{2\sqrt{5}}$

2.



Braille letter

Limit of a function, group *Chi* χ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 1/6} \frac{3}{(6x-1)^5}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 6^{2+4x^2}$??? converge

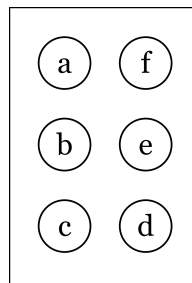
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x - 2} - \sqrt{x^2 - 3x - 3} \right)$??? -2

(d) $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x+3} \right)^{x/2}$??? e^2

(e) $\lim_{x \rightarrow -7} \frac{2x^2 + 11x - 21}{x^3 + 7x^2 + 5x + 35}$??? $-\frac{17}{108}$

(f) $\lim_{x \rightarrow 8} \frac{\sqrt{59} - \sqrt{7x+3}}{x-8}$??? 0

3.



Braille letter

Limit of a function, group *Chi* χ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 3} \frac{-1}{(-3x+9)^{17}}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{4+3x^2}$??? diverge

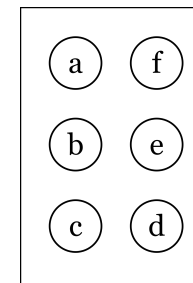
(c) $\lim_{x \rightarrow \infty} \left(3x - \sqrt{9x^2 - 6x - 6} \right)$??? 1

(d) $\lim_{x \rightarrow \infty} \left(\frac{-4x+1}{-4x-7} \right)^{x/6}$... ??? ... $e^{1/4}$

(e) $\lim_{x \rightarrow -2} \frac{5x^2 + 9x - 2}{-4x^3 - 8x^2 - 6x - 12}$??? $\frac{1}{2}$

(f) $\lim_{x \rightarrow 8} \frac{\sqrt{30} - \sqrt{4x-2}}{5x-40}$... ??? ... ∞

4.



Braille letter

Limit of a function, group *Psi* ψ -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 4/7} \frac{1}{(-7x+4)^7}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 4^{-3-x^2}$??? diverge

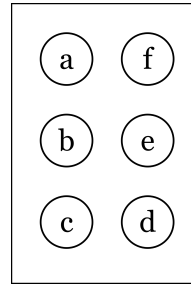
(c) $\lim_{x \rightarrow \infty} (8x - \sqrt{64x^2 - 6x + 9})$??? $\frac{3}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-x+5}{-x+3} \right)^{x/7}$... ??? ... $e^{-2/7}$

(e) $\lim_{x \rightarrow -1} \frac{2x^2 + 10x + 8}{-2x^3 - 2x^2 - 3x - 3}$. ??? . 0

(f) $\lim_{x \rightarrow -4} \frac{1 - \sqrt{x+5}}{4x+16}$??? ∞

1.



Braille letter

Limit of a function, group *Psi* ψ -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 5/4} \frac{3}{(4x-5)^{14}}$??? does exist

(b) $\lim_{x \rightarrow \infty} 7^{-4-6x^2}$??? diverge

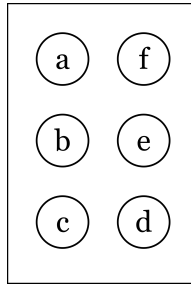
(c) $\lim_{x \rightarrow \infty} (\sqrt{36x^2 + 4x + 9} - \sqrt{36x^2 + 8x + 9})$??? $-\frac{1}{3}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x-1}{-5x+2} \right)^{x/4}$??? ∞

(e) $\lim_{x \rightarrow -3} \frac{7x^2 + 22x + 3}{-4x^3 - 12x^2 - 3x - 9}$??? $\frac{20}{39}$

(f) $\lim_{x \rightarrow 2} \frac{\sqrt{13} - \sqrt{8x-3}}{7x-14}$??? $\frac{-4}{7\sqrt{13}}$

2.



Braille letter

Limit of a function, group *Psi* ψ -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 2/3} \frac{-5}{(-9x+6)^3}$.. ??? .. does exist

(b) $\lim_{x \rightarrow \infty} 5^{1+4x^2}$??? diverge

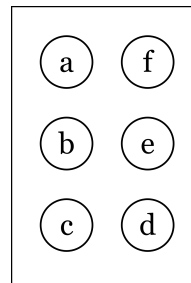
(c) $\lim_{x \rightarrow \infty} (4x - \sqrt{16x^2 - 3x + 1})$??? $-\frac{3}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-7x+3}{-7x+2} \right)^{x/6}$... ??? ... $e^{-5/42}$

(e) $\lim_{x \rightarrow -3} \frac{-5x^2 - 20x - 15}{-8x^3 - 24x^2 + 7x + 21}$??? $\frac{2}{65}$

(f) $\lim_{x \rightarrow 2} \frac{x-2}{\sqrt{2x-1} - \sqrt{1+x}}$.. ??? .. $\sqrt{12}$

3.



Braille letter

Limit of a function, group *Psi* ψ -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 3/4} \frac{-6}{(-4x+3)^8}$??? does not exist

(b) $\lim_{x \rightarrow \infty} 2^{-1-3x^2}$??? converge

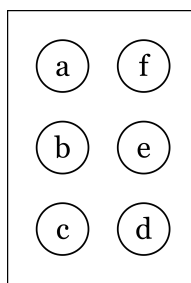
(c) $\lim_{x \rightarrow \infty} (3x - \sqrt{9x^2 + x - 8})$??? $-\frac{1}{6}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{x+3}{x-7} \right)^{x/7}$??? ∞

(e) $\lim_{x \rightarrow -2} \frac{2x^2 + 3x - 2}{x^3 + 2x^2 + 2x + 4}$. ??? . $-\frac{5}{6}$

(f) $\lim_{x \rightarrow 2/3} \frac{x - \frac{2}{3}}{\sqrt{3x+2} - \sqrt{6-3x}}$. ??? . $\frac{2}{3}$

4.



Braille letter

Limit of a function, group *Omega* ω -i

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow 2} \frac{-4}{(-2x+4)^9} \cdot ??? \cdot$ does not exist

(b) $\lim_{x \rightarrow \infty} 6^{-2+3x^2} \dots \dots \dots ??? \dots \dots \dots$ diverge

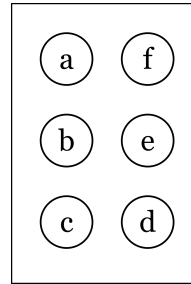
(c) $\lim_{x \rightarrow \infty} \left(4x - \sqrt{16x^2 - 9x + 9}\right) \quad ??? \quad -\frac{9}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-2x+7}{-2x-2}\right)^{x/2} \dots \dots \dots ??? \dots \dots \dots e^{-5/4}$

(e) $\lim_{x \rightarrow -4} \frac{2x^2 + 12x + 16}{x^3 + 4x^2 - 6x - 24} \cdot ??? \cdot \quad -\frac{2}{5}$

(f) $\lim_{x \rightarrow 1/2} \frac{x - \frac{1}{2}}{\sqrt{2x+3} - \sqrt{6-4x}} \dots \dots \dots ??? \dots \dots \dots 0$

1.



Braille letter

Limit of a function, group *Omega* ω -ii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -7/8} \frac{9}{(8x+7)^8} \dots \dots \dots ??? \dots \dots \dots$ does exist

(b) $\lim_{x \rightarrow \infty} 6^{4-3x^2} \dots \dots \dots ??? \dots \dots \dots$ diverge

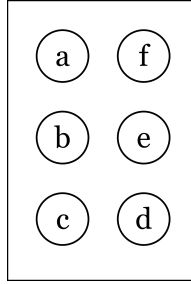
(c) $\lim_{x \rightarrow \infty} \left(5x - \sqrt{25x^2 - 5x - 1}\right) \quad ??? \quad -\frac{1}{2}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-5x+7}{-5x-4}\right)^{x/3} \dots \dots \dots ??? \dots \dots \dots e^{-1/5}$

(e) $\lim_{x \rightarrow -6} \frac{-3x^2 - 15x + 18}{x^3 + 6x^2 - 2x - 12} \dots \dots \dots ??? \dots \dots \dots \frac{21}{34}$

(f) $\lim_{x \rightarrow 5/2} \frac{4x-10}{\sqrt{3x-4} - \sqrt{1+x}} \dots \dots \dots ??? \dots \dots \dots 0$

2.



Braille letter

Limit of a function, group *Omega* ω -iii

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1} \frac{9}{(-3x-3)^3} \dots \dots \dots ??? \dots \dots \dots$ does not exist

(b) $\lim_{x \rightarrow \infty} 5^{7+5x^2} \dots \dots \dots ??? \dots \dots \dots$ converge

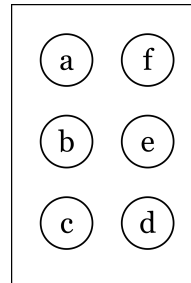
(c) $\lim_{x \rightarrow \infty} \left(\sqrt{16x^2 - 2x + 8} - \sqrt{16x^2 + x - 9}\right) \quad ??? \quad -\frac{1}{8}$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-7x-6}{-7x-5}\right)^{x/3} \dots \dots \dots ??? \dots \dots \dots e^{11/21}$

(e) $\lim_{x \rightarrow -3} \frac{3x^2 + 8x - 3}{x^2 - 9} \dots \dots \dots ??? \dots \dots \dots \frac{5}{9}$

(f) $\lim_{x \rightarrow 8} \frac{5 - \sqrt{3x+1}}{2x-16} \dots \dots \dots ??? \dots \dots \dots \infty$

3.



Braille letter

Limit of a function, group *Omega* ω -iv

(a) Determine whether the limit of the function exists.

(b) Determine whether the limit of the function converges or diverges.

(c) Evaluate the limit of the function.

(d) Evaluate the limit of the function.

(e) Evaluate the limit of the function.

(f) Evaluate the limit of the function.

(a) $\lim_{x \rightarrow -1} \frac{3}{(-x-1)^5} \cdot ??? \cdot$ does not exist

(b) $\lim_{x \rightarrow \infty} 3^{-1-x^2} \dots \dots \dots ??? \dots \dots \dots$ diverge

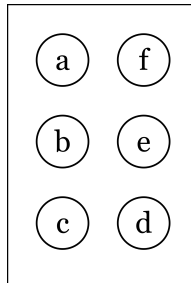
(c) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + 2x - 8}\right) \dots \dots \dots ??? \dots \dots \dots -2$

(d) $\lim_{x \rightarrow \infty} \left(\frac{-3x+5}{-3x-1}\right)^{x/5} \dots \dots \dots ??? \dots \dots \dots e^{-4/15}$

(e) $\lim_{x \rightarrow -4} \frac{-4x^2 - 22x - 24}{-3x^3 - 12x^2 + 3x + 12} \quad ??? \quad -\frac{2}{9}$

(f) $\lim_{x \rightarrow 1} \frac{2 - \sqrt{5x-1}}{4x-4} \dots \dots \dots ??? \dots \dots \dots -\frac{5}{16}$

4.



Braille letter

Limit of a function (solutions)

α	$i : \mathbf{W}$	(a) does not exist $\times$	(b) converge $\checkmark$	(c) $3/14 \times$	(d) $e^{-2/45} \checkmark$	(e) $3/4 \checkmark$	(f) $-1/12\sqrt{6} \checkmark$
	$ii : \mathbf{I}$	(a) does not exist $\times$	(b) converge $\checkmark$	(c) $-4/3 \times$	(d) $e^{1/14} \times$	(e) $13/3 \times$	(f) $-1/4 \checkmark$
	$iii : \mathbf{N}$	(a) does exist $\checkmark$	(b) converge $\times$	(c) $1 \checkmark$	(d) $e^{1/10} \times$	(e) $21/47 \checkmark$	(f) $12/\sqrt{125} \checkmark$
	$iv : \mathbf{D}$	(a) does exist $\checkmark$	(b) diverge $\times$	(c) $-5/4 \times$	(d) $e^{-3/5} \times$	(e) $1/2 \checkmark$	(f) $-7/18\sqrt{5} \checkmark$
β	$i : \mathbf{C}$	(a) does not exist $\checkmark$	(b) diverge $\times$	(c) $3 \times$	(d) $e^{-1/12} \times$	(e) $-2 \times$	(f) $-2/\sqrt{11} \checkmark$
	$ii : \mathbf{H}$	(a) does not exist $\checkmark$	(b) converge $\checkmark$	(c) $4 \times$	(d) $e^4 \times$	(e) $-12/7 \checkmark$	(f) $20/\sqrt{343} \times$
	$iii : \mathbf{A}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $-3/8 \times$	(d) $e^{-1/2} \times$	(e) $5/9 \times$	(f) $-1/\sqrt{18} \times$
	$iv : \mathbf{T}$	(a) does exist $\times$	(b) converge $\checkmark$	(c) $-1/7 \checkmark$	(d) $e^{-1/5} \times$	(e) $-2/13 \checkmark$	(f) $-3/4\sqrt{17} \checkmark$
γ	$i : \mathbf{H}$	(a) does exist $\checkmark$	(b) diverge $\checkmark$	(c) $-3/2 \times$	(d) $e^{-1/3} \times$	(e) $7/6 \checkmark$	(f) $\sqrt{20}/\sqrt{343} \times$
	$ii : \mathbf{O}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $3/5 \checkmark$	(d) $e^{2/3} \times$	(e) $-15/8 \checkmark$	(f) $-7/12\sqrt{58} \times$
	$iii : \mathbf{P}$	(a) does exist $\checkmark$	(b) converge $\checkmark$	(c) $3/8 \checkmark$	(d) $e^{7/27} \times$	(e) $7/2 \times$	(f) $-7/6\sqrt{21} \checkmark$
	$iv : \mathbf{E}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $-3/2 \times$	(d) $e^{-2/21} \times$	(e) $1/2 \checkmark$	(f) $\sqrt{368}/\sqrt{125} \times$
δ	$i : \mathbf{H}$	(a) does exist $\checkmark$	(b) diverge $\checkmark$	(c) $3/16 \times$	(d) $e^{17/24} \times$	(e) $-7/30 \checkmark$	(f) $\sqrt{6} \times$
	$ii : \mathbf{A}$	(a) does exist $\checkmark$	(b) diverge $\times$	(c) $7/2 \times$	(d) $e^{1/24} \times$	(e) $-5/28 \times$	(f) $-7/18\sqrt{13} \times$
	$iii : \mathbf{N}$	(a) does exist $\checkmark$	(b) converge $\times$	(c) $-13/16 \checkmark$	(d) $e^{-3/20} \times$	(e) $-8/37 \checkmark$	(f) $-5/24 \checkmark$
	$iv : \mathbf{D}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $-3/2 \times$	(d) $e^{-7/4} \times$	(e) $-10/79 \checkmark$	(f) $\sqrt{14}/\sqrt{125} \checkmark$
ϵ	$i : \mathbf{S}$	(a) does exist $\times$	(b) diverge $\checkmark$	(c) $-3/4 \checkmark$	(d) $e^{-3/2} \times$	(e) $3 \times$	(f) $1/\sqrt{32} \checkmark$
	$ii : \mathbf{I}$	(a) does exist $\times$	(b) converge $\checkmark$	(c) $7/12 \times$	(d) $e^{1/4} \times$	(e) $-17/4 \times$	(f) $8/\sqrt{27} \checkmark$
	$iii : \mathbf{Z}$	(a) does not exist $\checkmark$	(b) diverge $\times$	(c) $-1/16 \checkmark$	(d) $e^{1/6} \checkmark$	(e) $11/3 \checkmark$	(f) $-1/10\sqrt{7} \times$
	$iv : \mathbf{E}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $1/16 \times$	(d) $e^{3/8} \times$	(e) $15/74 \checkmark$	(f) $18/\sqrt{125} \times$
ζ	$i : \mathbf{C}$	(a) does exist $\checkmark$	(b) diverge $\times$	(c) $1/8 \times$	(d) $e^{5/12} \times$	(e) $-3/4 \times$	(f) $\sqrt{8} \checkmark$
	$ii : \mathbf{I}$	(a) does not exist $\times$	(b) converge $\checkmark$	(c) $5/4 \times$	(d) $e^{5/2} \times$	(e) $-19/70 \times$	(f) $-5/24 \checkmark$
	$iii : \mathbf{T}$	(a) does exist $\times$	(b) diverge $\checkmark$	(c) $1 \checkmark$	(d) $e^{3/2} \times$	(e) $-10/3 \checkmark$	(f) $\sqrt{3} \checkmark$
	$iv : \mathbf{Y}$	(a) does exist $\checkmark$	(b) converge $\times$	(c) $-1/3 \checkmark$	(d) $e^{11/14} \checkmark$	(e) $7/6 \checkmark$	(f) $-1/5\sqrt{10} \checkmark$
η	$i : \mathbf{D}$	(a) does exist $\checkmark$	(b) converge $\times$	(c) $-2/5 \times$	(d) $e^{-13/35} \times$	(e) $-13/44 \checkmark$	(f) $-3/4\sqrt{32} \checkmark$
	$ii : \mathbf{A}$	(a) does exist $\checkmark$	(b) converge $\times$	(c) $-5/6 \times$	(d) $e^{-1/5} \times$	(e) $-1/4 \times$	(f) $-1/6\sqrt{2} \times$
	$iii : \mathbf{T}$	(a) does exist $\times$	(b) converge $\checkmark$	(c) $3/5 \checkmark$	(d) $e^{-9/12} \times$	(e) $-15 \checkmark$	(f) $-5/18\sqrt{5} \checkmark$
	$iv : \mathbf{E}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $1/4 \times$	(d) $e^{-1/27} \times$	(e) $-12/5 \checkmark$	(f) $-3/8\sqrt{6} \times$
θ	$i : \mathbf{M}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $5/12 \checkmark$	(d) $e^2 \times$	(e) $2/3 \times$	(f) $-1/4\sqrt{5} \checkmark$
	$ii : \mathbf{I}$	(a) does not exist $\times$	(b) diverge $\checkmark$	(c) $1 \times$	(d) $e^{9/12} \times$	(e) $-13/47 \times$	(f) $-1/10\sqrt{11} \checkmark$
	$iii : \mathbf{N}$	(a) does exist $\checkmark$	(b) diverge $\times$	(c) $3/16 \checkmark$	(d) $e^{11/6} \times$	(e) $19/4 \checkmark$	(f) $\sqrt{112}/\sqrt{27} \checkmark$
	$iv : \mathbf{D}$	(a) does not exist $\checkmark$	(b) diverge $\times$	(c) $1/2 \times$	(d) $e^{-1/18} \times$	(e) $5/2 \checkmark$	(f) $-1/2\sqrt{5} \checkmark$
ι	$i : \mathbf{B}$	(a) does exist $\checkmark$	(b) converge $\checkmark$	(c) $-1/10 \times$	(d) $e^{-2/21} \times$	(e) $23/6 \times$	(f) $\sqrt{112} \times$
	$ii : \mathbf{A}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $1/8 \times$	(d) $e^{1/4} \times$	(e) $3/2 \times$	(f) $-3/2\sqrt{7} \times$
	$iii : \mathbf{L}$	(a) does exist $\checkmark$	(b) diverge $\checkmark$	(c) $1/2 \checkmark$	(d) $e^{-1/4} \times$	(e) $-22/83 \times$	(f) $\sqrt{40}/\sqrt{27} \times$
	$iv : \mathbf{L}$	(a) does not exist $\checkmark$	(b) diverge $\checkmark$	(c) $-1/3 \checkmark$	(d) $e^{-12/25} \times$	(e) $3 \times$	(f) $-3/4\sqrt{19} \times$
κ	$i : \mathbf{N}$	(a) does exist $\checkmark$	(b) diverge $\times$	(c) $2/5 \checkmark$	(d) $e^{1/12} \times$	(e) $5 \checkmark$	(f) $\sqrt{8}/\sqrt{125} \checkmark$
	$ii : \mathbf{E}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $1/2 \times$	(d) $e^{-1/28} \times$	(e) $-1 \checkmark$	(f) $-3/4\sqrt{3} \times$
	$iii : \mathbf{W}$	(a) does exist $\times$	(b) diverge $\checkmark$	(c) $-5/2 \times$	(d) $e^{1/8} \checkmark$	(e) $-1 \checkmark$	(f) $1/\sqrt{18} \checkmark$
	$iv : \mathbf{S}$	(a) does exist $\times$	(b) converge $\checkmark$	(c) $-6/7 \checkmark$	(d) $e^{1/3} \times$	(e) $-13/8 \times$	(f) $-3/2 \checkmark$
λ	$i : \mathbf{H}$	(a) does exist $\checkmark$	(b) diverge $\checkmark$	(c) $-1/2 \times$	(d) $e^{-1/6} \times$	(e) $7/2 \checkmark$	(f) $-3/7\sqrt{33} \times$
	$ii : \mathbf{O}$	(a) does not exist $\checkmark$	(b) converge $\times$	(c) $-5/2 \checkmark$	(d) $e^{1/2} \times$	(e) $4 \checkmark$	(f) $-1/6\sqrt{11} \times$
	$iii : \mathbf{M}$	(a) does exist $\checkmark$	(b) diverge $\times$	(c) $-2/3 \checkmark$	(d) $e^{-1/2} \times$	(e) $47/14 \times$	(f) $-9/4\sqrt{59} \checkmark$
	$iv : \mathbf{E}$	(a) does exist $\checkmark$	(b) converge $\times$	(c) $1/6 \times$	(d) $e^{2/3} \times$	(e) $-7/2 \checkmark$	(f) $-7/8 \times$
μ	$i : \mathbf{W}$	(a) does not exist $\times$	(b) converge $\checkmark$	(c) $-1/2 \times$	(d) $e^{13/24} \checkmark$	(e) $-1/2 \checkmark$	(f) $\sqrt{11}/\sqrt{27} \checkmark$
	$ii : \mathbf{I}$	(a) does exist $\times$	(b) diverge $\checkmark$	(c) $-1/16 \times$	(d) $e^{2/5} \times$	(e) $-1/2 \times$	(f) $\sqrt{32}/\sqrt{3} \checkmark$
	$iii : \mathbf{S}$	(a) does not exist $\times$	(b) converge $\checkmark$	(c) $1/16 \checkmark$	(d) $e^{-1} \times$	(e) $-1/6 \times$	(f) $\sqrt{112}/\sqrt{27} \checkmark$
	$iv : \mathbf{H}$	(a) does exist $\checkmark$	(b) diverge $\checkmark$	(c) $4/3 \times$	(d) $e^{-1/3} \times$	(e) $35/8 \checkmark$	(f) $-1 \times$

Limit of a function (solutions)

ν	$i : \mathbf{N}$	(a) does exist ✓	(b) diverge $\mathbf{X}$	(c) $-1/3$ ✓	(d) $e^{13/16} \mathbf{X}$	(e) $11/2$ ✓	(f) $-1/14\sqrt{10}$ ✓
	$ii : \mathbf{A}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $-1/4 \mathbf{X}$	(d) $e^{3/2} \mathbf{X}$	(e) $-8/33 \mathbf{X}$	(f) $-3/2\sqrt{25} \mathbf{X}$
	$iii : \mathbf{M}$	(a) does not exist ✓	(b) converge $\mathbf{X}$	(c) -1 ✓	(d) $e^{-1/10} \mathbf{X}$	(e) $1/3 \mathbf{X}$	(f) $\sqrt{700}/\sqrt{27}$ ✓
	$iv : \mathbf{E}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $-3/14 \mathbf{X}$	(d) $e^{2/3} \mathbf{X}$	(e) $-34/45$ ✓	(f) $8 \mathbf{X}$
ξ	$i : \mathbf{I}$	(a) does exist $\mathbf{X}$	(b) diverge ✓	(c) $-3/8 \mathbf{X}$	(d) $e^{-1/2} \mathbf{X}$	(e) $-38/27 \mathbf{X}$	(f) $-1/2\sqrt{6}$ ✓
	$ii : \mathbf{D}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $-2 \mathbf{X}$	(d) $e^{-5/8} \mathbf{X}$	(e) -2 ✓	(f) $\sqrt{168}/\sqrt{125}$ ✓
	$iii : \mathbf{E}$	(a) does exist ✓	(b) diverge $\mathbf{X}$	(c) $-7/6 \mathbf{X}$	(d) $e^{-5/18} \mathbf{X}$	(e) $-8/3$ ✓	(f) $8/\sqrt{27} \mathbf{X}$
	$iv : \mathbf{A}$	(a) does not exist ✓	(b) converge $\mathbf{X}$	(c) $1 \mathbf{X}$	(d) $e^{-1/4} \mathbf{X}$	(e) $-15/2 \mathbf{X}$	(f) $\sqrt{3}/\sqrt{8} \mathbf{X}$
σ	$i : \mathbf{C}$	(a) does not exist ✓	(b) diverge $\mathbf{X}$	(c) $-7/2 \mathbf{X}$	(d) $e^{-1/2} \mathbf{X}$	(e) $7/50 \mathbf{X}$	(f) $-1/14\sqrt{5}$ ✓
	$ii : \mathbf{A}$	(a) does not exist ✓	(b) converge $\mathbf{X}$	(c) $-1/4 \mathbf{X}$	(d) $e^{1/2} \mathbf{X}$	(e) $-3/20 \mathbf{X}$	(f) $\sqrt{28}/\sqrt{3} \mathbf{X}$
	$iii : \mathbf{S}$	(a) does not exist $\mathbf{X}$	(b) diverge ✓	(c) $-9/16$ ✓	(d) $e^{-1} \mathbf{X}$	(e) $2/5 \mathbf{X}$	(f) $-5/8\sqrt{32}$ ✓
	$iv : \mathbf{E}$	(a) does not exist ✓	(b) diverge $\mathbf{X}$	(c) $2/5 \mathbf{X}$	(d) $e^{2/3} \mathbf{X}$	(e) 6 ✓	(f) $\sqrt{112}/\sqrt{125} \mathbf{X}$
π	$i : \mathbf{F}$	(a) does not exist ✓	(b) converge ✓	(c) $1/16 \mathbf{X}$	(d) $e^{9/35} \mathbf{X}$	(e) $-1 \mathbf{X}$	(f) $2/\sqrt{125}$ ✓
	$ii : \mathbf{O}$	(a) does not exist ✓	(b) diverge $\mathbf{X}$	(c) $1/3$ ✓	(d) $e^{-1/27} \mathbf{X}$	(e) $3/2$ ✓	(f) $\sqrt{68}/\sqrt{27} \mathbf{X}$
	$iii : \mathbf{O}$	(a) does not exist ✓	(b) converge $\mathbf{X}$	(c) $-1/4$ ✓	(d) $e^{-7/15} \mathbf{X}$	(e) $-3/2$ ✓	(f) $\sqrt{12}/\sqrt{125} \mathbf{X}$
	$iv : \mathbf{T}$	(a) does not exist $\mathbf{X}$	(b) diverge ✓	(c) $-3/2$ ✓	(d) $e^{1/5} \mathbf{X}$	(e) $11/3$ ✓	(f) $-1/\sqrt{20}$ ✓
ρ	$i : \mathbf{A}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $2 \mathbf{X}$	(d) $e^{-3/4} \mathbf{X}$	(e) $-19/25 \mathbf{X}$	(f) $-5/2\sqrt{41} \mathbf{X}$
	$ii : \mathbf{C}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $-4 \mathbf{X}$	(d) $e^{-3/28} \mathbf{X}$	(e) $2/3 \mathbf{X}$	(f) $-2/\sqrt{27}$ ✓
	$iii : \mathbf{I}$	(a) does exist $\mathbf{X}$	(b) converge ✓	(c) $-1/6 \mathbf{X}$	(d) $e^{-3/4} \mathbf{X}$	(e) $-37/24 \mathbf{X}$	(f) $\sqrt{192}/\sqrt{125}$ ✓
	$iv : \mathbf{D}$	(a) does not exist ✓	(b) converge $\mathbf{X}$	(c) $-5/12 \mathbf{X}$	(d) $e^{-9/24} \mathbf{X}$	(e) $18/71$ ✓	(f) $-4/3\sqrt{11}$ ✓
σ	$i : \mathbf{B}$	(a) does exist ✓	(b) converge ✓	(c) $6 \mathbf{X}$	(d) $e^{-1/2} \mathbf{X}$	(e) $15/44 \mathbf{X}$	(f) $\sqrt{96} \mathbf{X}$
	$ii : \mathbf{O}$	(a) does exist ✓	(b) diverge $\mathbf{X}$	(c) $1/2$ ✓	(d) $e^{1/30} \mathbf{X}$	(e) $5/2$ ✓	(f) $-1/7\sqrt{12} \mathbf{X}$
	$iii : \mathbf{S}$	(a) does not exist $\mathbf{X}$	(b) converge ✓	(c) -3 ✓	(d) $e^{-7/12} \mathbf{X}$	(e) $1/3 \mathbf{X}$	(f) $-5/2\sqrt{43}$ ✓
	$iv : \mathbf{S}$	(a) does exist $\mathbf{X}$	(b) diverge ✓	(c) $-7/4$ ✓	(d) $e^{-1/5} \mathbf{X}$	(e) $15/34 \mathbf{X}$	(f) $\sqrt{7}/\sqrt{3}$ ✓
τ	$i : \mathbf{P}$	(a) does not exist ✓	(b) diverge ✓	(c) $13/2$ ✓	(d) $e^{-2} \mathbf{X}$	(e) $-5/3 \mathbf{X}$	(f) $-7/4\sqrt{24}$ ✓
	$ii : \mathbf{L}$	(a) does not exist ✓	(b) diverge ✓	(c) $-1/6$ ✓	(d) $e^{-5/16} \mathbf{X}$	(e) $17/5 \mathbf{X}$	(f) $\sqrt{72}/\sqrt{343} \mathbf{X}$
	$iii : \mathbf{A}$	(a) does exist ✓	(b) diverge $\mathbf{X}$	(c) $1/8 \mathbf{X}$	(d) $e^{-9/4} \mathbf{X}$	(e) $7/2 \mathbf{X}$	(f) $-1/8 \mathbf{X}$
	$iv : \mathbf{N}$	(a) does not exist ✓	(b) converge $\mathbf{X}$	(c) $3/2$ ✓	(d) $e^{11/42} \mathbf{X}$	(e) $-13/5$ ✓	(f) $2/7$ ✓
v	$i : \mathbf{S}$	(a) does not exist $\mathbf{X}$	(b) diverge ✓	(c) 0 ✓	(d) $e^{2/7} \mathbf{X}$	(e) $-1/12 \mathbf{X}$	(f) $\sqrt{352}$ ✓
	$ii : \mathbf{A}$	(a) does exist ✓	(b) diverge $\mathbf{X}$	(c) $-2/7 \mathbf{X}$	(d) $e^2 \mathbf{X}$	(e) $-19/8 \mathbf{X}$	(f) $-2/5\sqrt{7} \mathbf{X}$
	$iii : \mathbf{L}$	(a) does not exist ✓	(b) converge ✓	(c) $3/8$ ✓	(d) $e^{7/24} \mathbf{X}$	(e) $-5/4 \mathbf{X}$	(f) $-1/6 \mathbf{X}$
	$iv : \mathbf{E}$	(a) does not exist ✓	(b) diverge $\mathbf{X}$	(c) $1/9 \mathbf{X}$	(d) $e^{-1/5} \mathbf{X}$	(e) $-5/2$ ✓	(f) $-3/2\sqrt{77} \mathbf{X}$
ϕ	$i : \mathbf{A}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $-2/3 \mathbf{X}$	(d) $e^{2/25} \mathbf{X}$	(e) $1/4 \mathbf{X}$	(f) $\sqrt{5}/\sqrt{8} \mathbf{X}$
	$ii : \mathbf{T}$	(a) does not exist $\mathbf{X}$	(b) converge ✓	(c) $3/10$ ✓	(d) $e^{-1/15} \mathbf{X}$	(e) $-10/3$ ✓	(f) $-2/\sqrt{21}$ ✓
	$iii : \mathbf{O}$	(a) does not exist ✓	(b) diverge $\mathbf{X}$	(c) $1/2$ ✓	(d) $e^{1/3} \mathbf{X}$	(e) $-1/26$ ✓	(f) $\sqrt{48}/\sqrt{125} \mathbf{X}$
	$iv : \mathbf{M}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $-1/12$ ✓	(d) $e^{3/5} \mathbf{X}$	(e) $40/53 \mathbf{X}$	(f) $-2/\sqrt{13}$ ✓
χ	$i : \mathbf{S}$	(a) does exist $\mathbf{X}$	(b) diverge ✓	(c) $-5/12$ ✓	(d) $e^{-3/5} \mathbf{X}$	(e) $-5/3 \mathbf{X}$	(f) $\sqrt{52}/\sqrt{125}$ ✓
	$ii : \mathbf{T}$	(a) does not exist $\mathbf{X}$	(b) diverge ✓	(c) -3 ✓	(d) $e^{-1/16} \mathbf{X}$	(e) $11/3$ ✓	(f) $-1/2\sqrt{5}$ ✓
	$iii : \mathbf{A}$	(a) does not exist ✓	(b) diverge $\mathbf{X}$	(c) $2 \mathbf{X}$	(d) $e^{-1} \mathbf{X}$	(e) $-17/54 \mathbf{X}$	(f) $-7/2\sqrt{59} \mathbf{X}$
	$iv : \mathbf{R}$	(a) does not exist ✓	(b) diverge ✓	(c) 1 ✓	(d) $e^{-1/3} \mathbf{X}$	(e) $1/2$ ✓	(f) $-2/5\sqrt{30} \mathbf{X}$
ψ	$i : \mathbf{U}$	(a) does not exist ✓	(b) converge $\mathbf{X}$	(c) $3/8$ ✓	(d) $e^{-2/7} \mathbf{X}$	(e) $-6/5 \mathbf{X}$	(f) $-1/8 \mathbf{X}$
	$ii : \mathbf{N}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $-1/3$ ✓	(d) $e^{3/20} \mathbf{X}$	(e) $20/39$ ✓	(f) $-4/7\sqrt{13}$ ✓
	$iii : \mathbf{I}$	(a) does not exist $\mathbf{X}$	(b) diverge ✓	(c) $3/8 \mathbf{X}$	(d) $e^{-1/42} \mathbf{X}$	(e) $-2/13 \mathbf{X}$	(f) $\sqrt{12}$ ✓
	$iv : \mathbf{T}$	(a) does exist $\mathbf{X}$	(b) converge ✓	(c) $-1/6$ ✓	(d) $e^{10/7} \mathbf{X}$	(e) $-5/6$ ✓	(f) $2/3$ ✓
ω	$i : \mathbf{H}$	(a) does not exist ✓	(b) diverge ✓	(c) $9/8 \mathbf{X}$	(d) $e^{-9/4} \mathbf{X}$	(e) $-2/5$ ✓	(f) $2/3 \mathbf{X}$
	$ii : \mathbf{E}$	(a) does exist ✓	(b) converge $\mathbf{X}$	(c) $1/2 \mathbf{X}$	(d) $e^{-11/15} \mathbf{X}$	(e) $21/34$ ✓	(f) $\sqrt{56} \mathbf{X}$
	$iii : \mathbf{A}$	(a) does not exist ✓	(b) diverge $\mathbf{X}$	(c) $-3/8 \mathbf{X}$	(d) $e^{1/21} \mathbf{X}$	(e) $5/3 \mathbf{X}$	(f) $-3/20 \mathbf{X}$
	$iv : \mathbf{D}$	(a) does not exist ✓	(b) converge $\mathbf{X}$	(c) $-1 \mathbf{X}$	(d) $e^{-2/5} \mathbf{X}$	(e) $-2/9$ ✓	(f) $-5/16$ ✓